Cloud-Native Geospatial

Standards for reproducible and scalable science

Matthew Hanson
@GeoSkeptic

NORTH51 2022, Banff

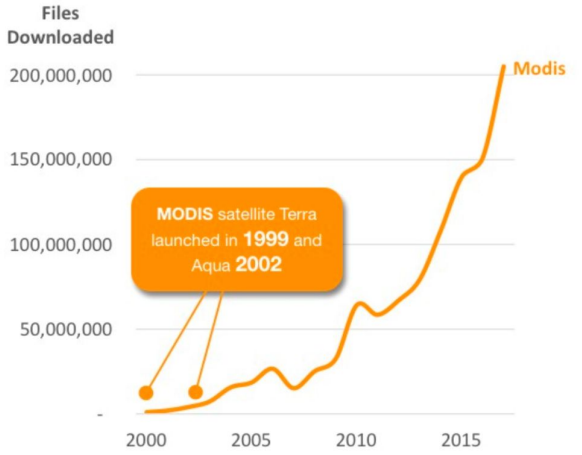
Element 84

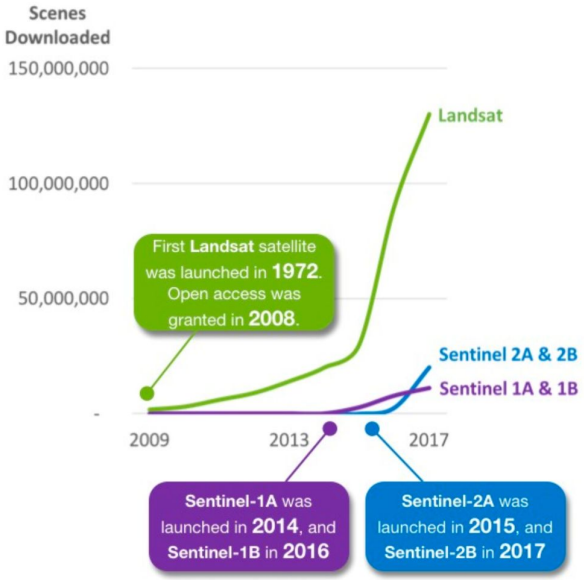
Standards seem boring

Analysis Ready Data,
a cautionary tale

Open Satellite Data Downloads

The number of downloads every year from the open data portals has increased exponentially!





SOURCES: MODIS and Landsat data was provided by NASA
Sentinel data was obtained via their annual reports, 2015-2017

*Files downloaded from the MODIS portal often contain multiple scenes

Radiant.Earth
Earth Imagery for Impact

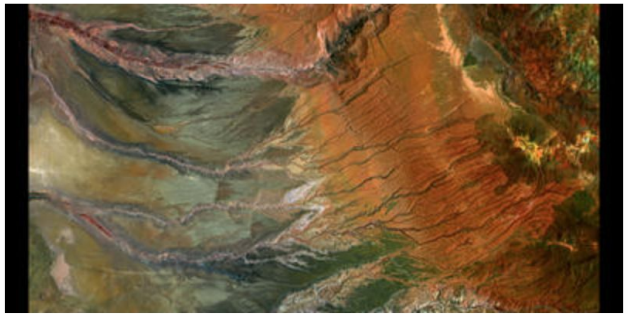
CC BY

Landsat 8 Data Users Handbook

The Landsat Data Users Handbooks are prepared by the U.S. Geological Survey (USGS) Landsat Project Science Office at the Earth Resources Observation and Science (EROS) Center in Sioux Falls, SD, and the National Aeronautics and Space Administration (NASA) Landsat Project Science Office at NASA's Goddard Space Flight Center (GSFC) in Greenbelt, Maryland.

Access Landsat 7 Data User Handbook

Download Landsat 8 Handbook



Download PDF file

5.2 OLI Top of Atmosphere Reflectance

Similar to the conversion to radiance, the 16-bit integer values in the Level 1 product can also be converted to TOA reflectance. The following equation is used to convert Level 1 DN values to TOA reflectance:

$$\rho\lambda' = M_\rho * Q_{cal} + A_\rho$$

where:

$\rho\lambda'$ = TOA Planetary Spectral Reflectance, without correction for solar angle. (Unitless)
M_ρ = Reflectance multiplicative scaling factor for the band (REFLECTANCEW_MULT_BAND_n from the metadata).
A_ρ = Reflectance additive scaling factor for the band (REFLECTANCE_ADD_BAND_N from the metadata).
Q_{cal} = Level 1 pixel value in DN

Note that $\rho\lambda'$ is not true TOA Reflectance, because it does not contain a correction for the solar elevation angle. This correction factor is left out of the Level 1 scaling at the users' request; some users are content with the scene-center solar elevation angle in the metadata, while others prefer to calculate their own per-pixel solar elevation angle across the entire scene. Once a solar elevation angle is chosen, the conversion to true TOA Reflectance is as follows:

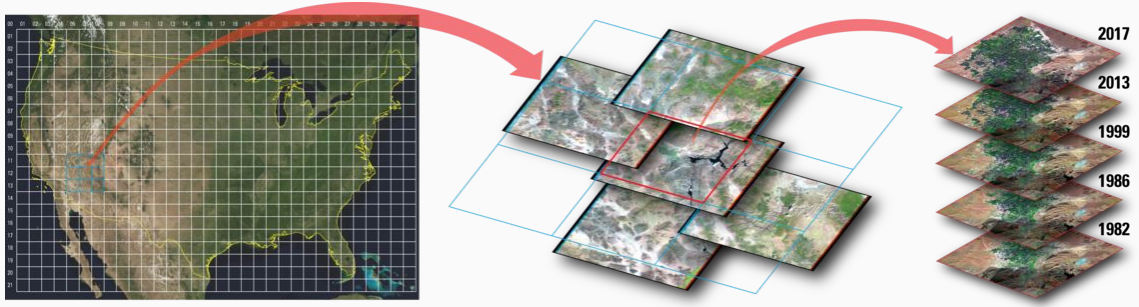
$$\rho_\lambda = \frac{\rho_\lambda'}{\cos(\theta_{SZ})} = \frac{\rho_\lambda'}{\sin(\theta_{SE})}$$

Landsat, the Original ARD

"The ARD are available to significantly reduce the burden of pre-processing on users of Landsat data."

Goal: Treat all Landsat as a "virtual" constellation to enable easy time series analysis across all platforms

- Processing
  - Surface reflectance (Atmospherically corrected)
  - Georegistered
- Metadata
  - Cloud/QC mask
- Formatting
  - Tiled
  - Common projection



ARD means easy discovery, access, and exploitation of measured geospatial data

What is Cloud-Native Geospatial?

It’s Google Earth Engine

A Queryable Earth

- How many ships are in all ports in the US?
- How many swimming pools were built in 2019 in Texas?
- How many acres of forest burned in California over the last year?
- What is the average size home in Alberta?
- Total km of roads in India?
- What is the electrification rate in Malawi?

Cloud-Native Geo means easy discovery, access, and exploitation of measured geospatial data (in the cloud)

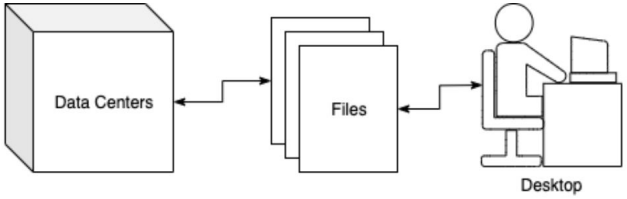
Except GEE is not..

- Open-Source
- based on standards
- deployable
- Easy to integrate with commercial providers
- Easily extensible via services

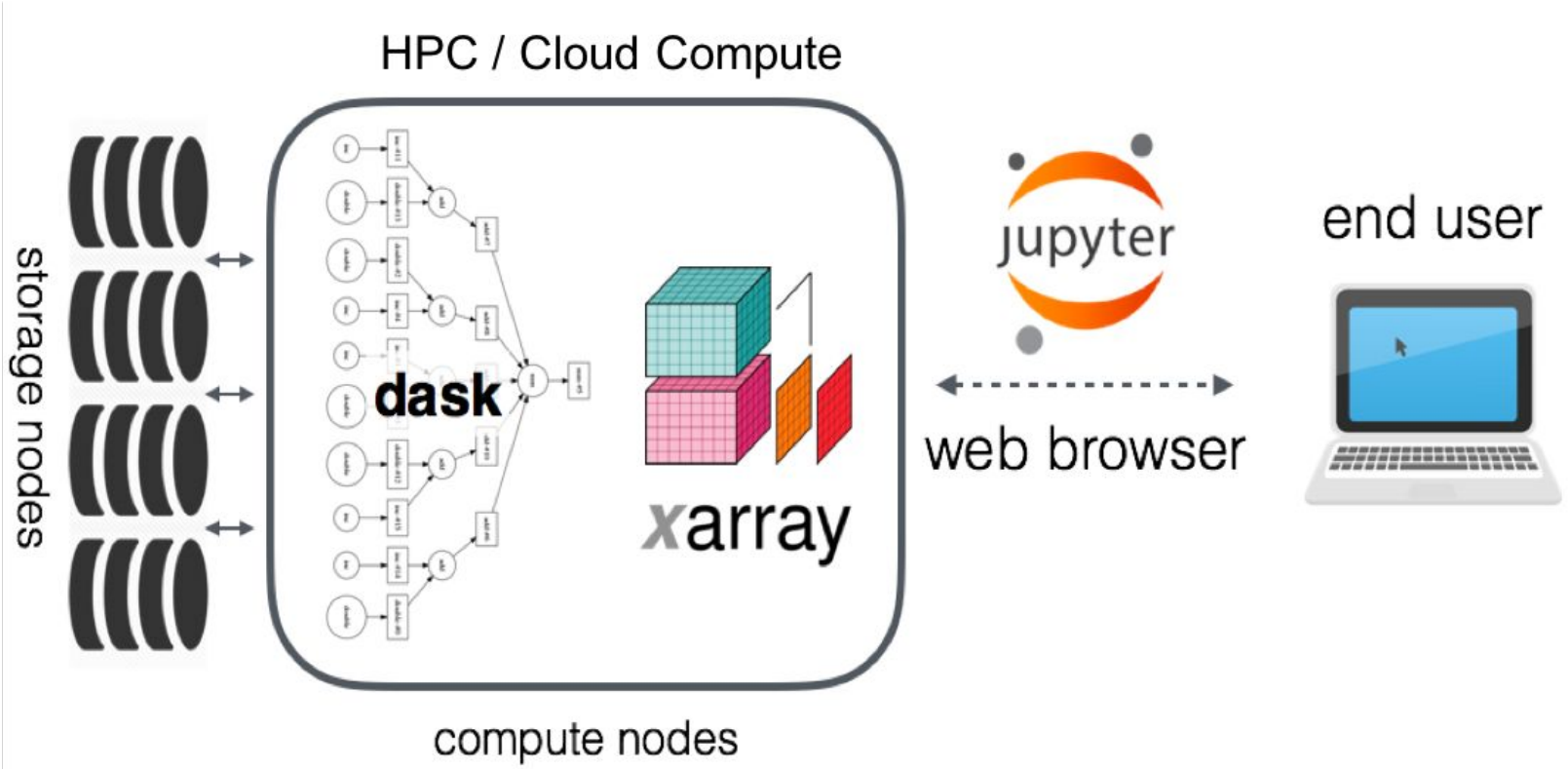
Cloud-Native Geo is the Foundational Architecture for a Queryable Earth

Cloud-Native Geospatial

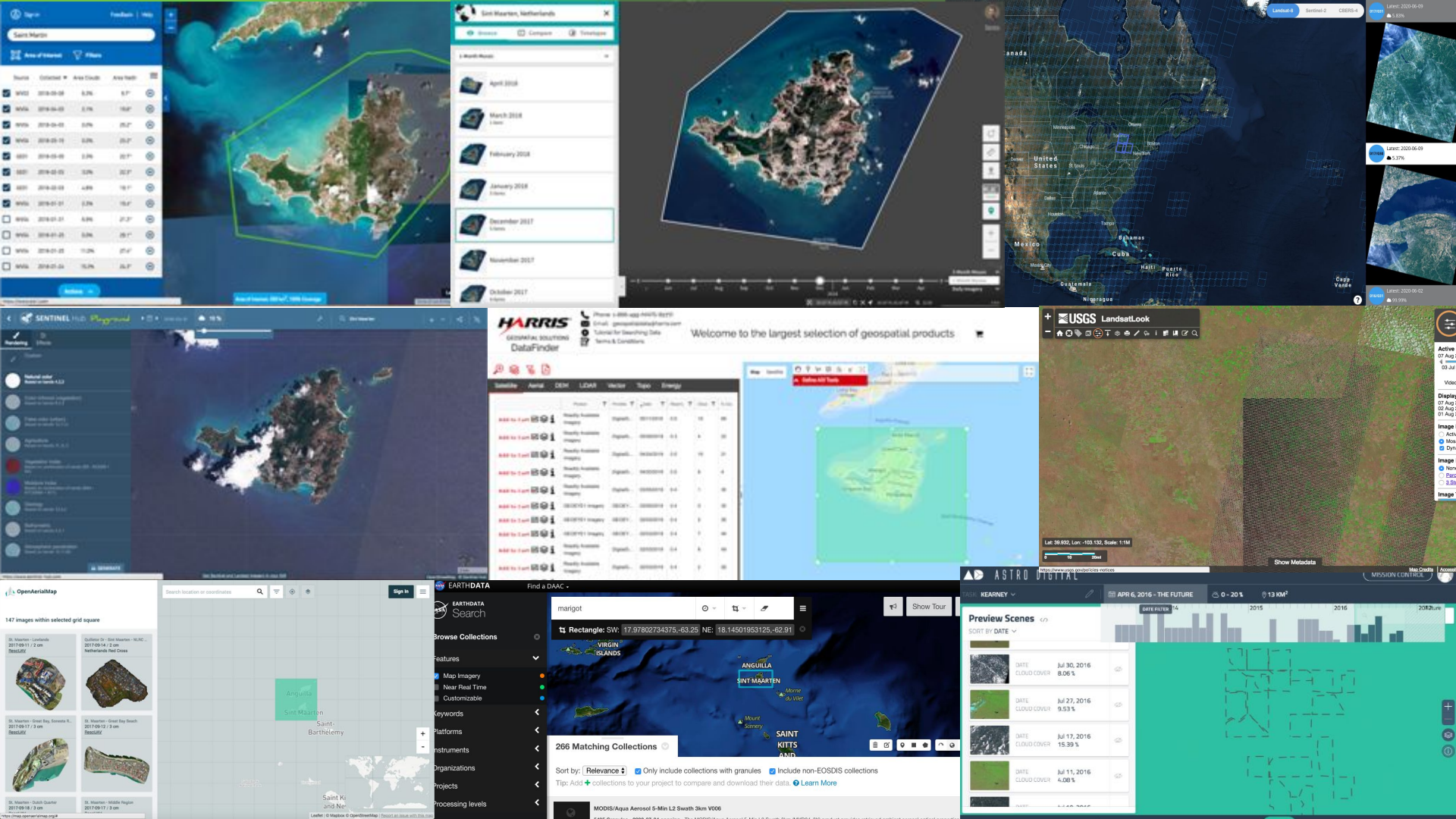
- Complete workflows performed in the cloud
  - Data Discovery
  - Processing
  - Analytics
  - Visualization
- via direct data access
  - By users OR services
- which promotes
  - Scalability
  - Reproducibility (provenance)
  - Online Publishing (executable papers)



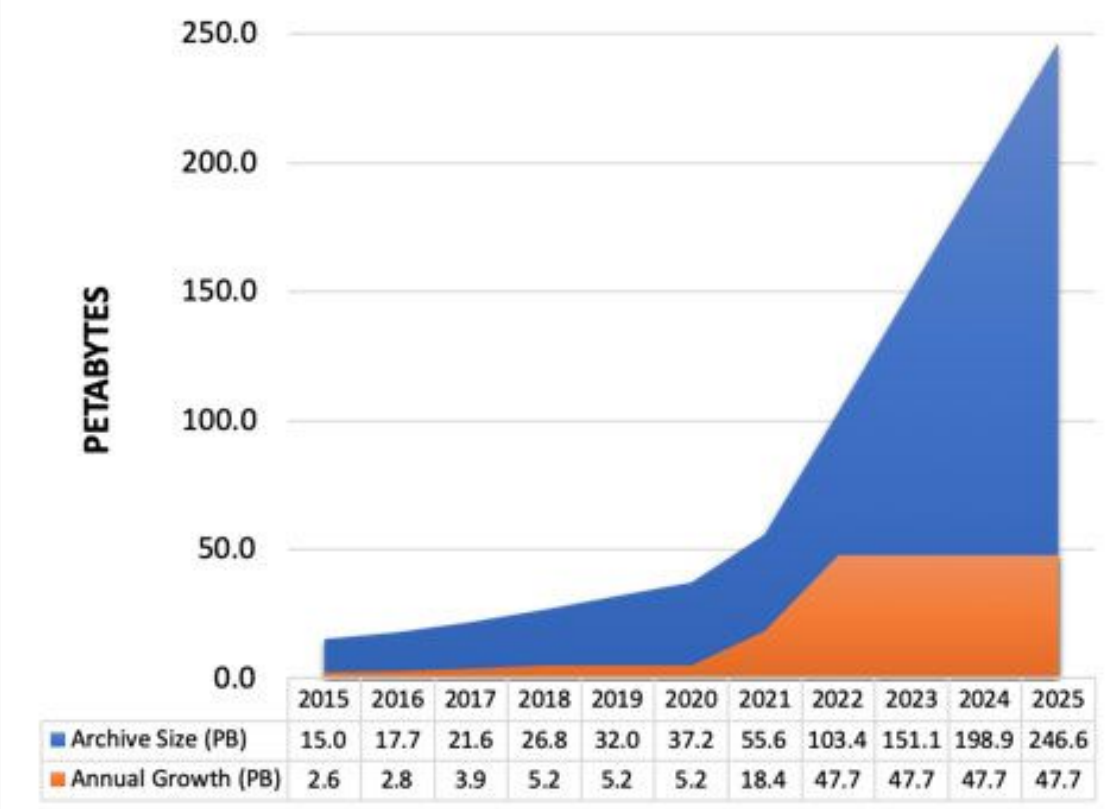
Pangeo


HPC / Cloud Compute
storage nodes
dask
xarray
compute nodes
jupyter
web browser
end user

SpatioTemporal Asset Catalogs (STAC)


Saint Martin
Sint Maarten, Netherlands
Welcome to the largest selection of geospatial products
HARRIS GEOSPATIAL SOLUTIONS DataFinder
USGS LandsatLook
EARTHDATA Search
Find a DAAC
marigot
Show Tour
Rectangle: SW: 17.97802734375,-63.25 NE: 18.14501953125,-62.91
VIRGIN ISLANDS
ANGUILLA
SINT MAARTEN
SAINT KITTS AND
266 Matching Collections
Sort by: Relevance
Only include collections with granules
Include non-EOSDIS collections
MODIS/Aqua Aerosol 5-Min L2 Swath 3km V006
ASTRO DIGITAL
KEARNEY
APR 6, 2016 - THE FUTURE
0 - 20 %
13 KM²
Preview Scenes
SORT BY DATE
Jul 30, 2016
8.06 %
Jul 27, 2016
9.93 %
Jul 17, 2016
15.39 %
Jul 11, 2016
4.08 %
OpenAerialMap
147 images within selected grid square
Anguilla
Sint Maarten
Saint-Barthelemy
Show Metadata

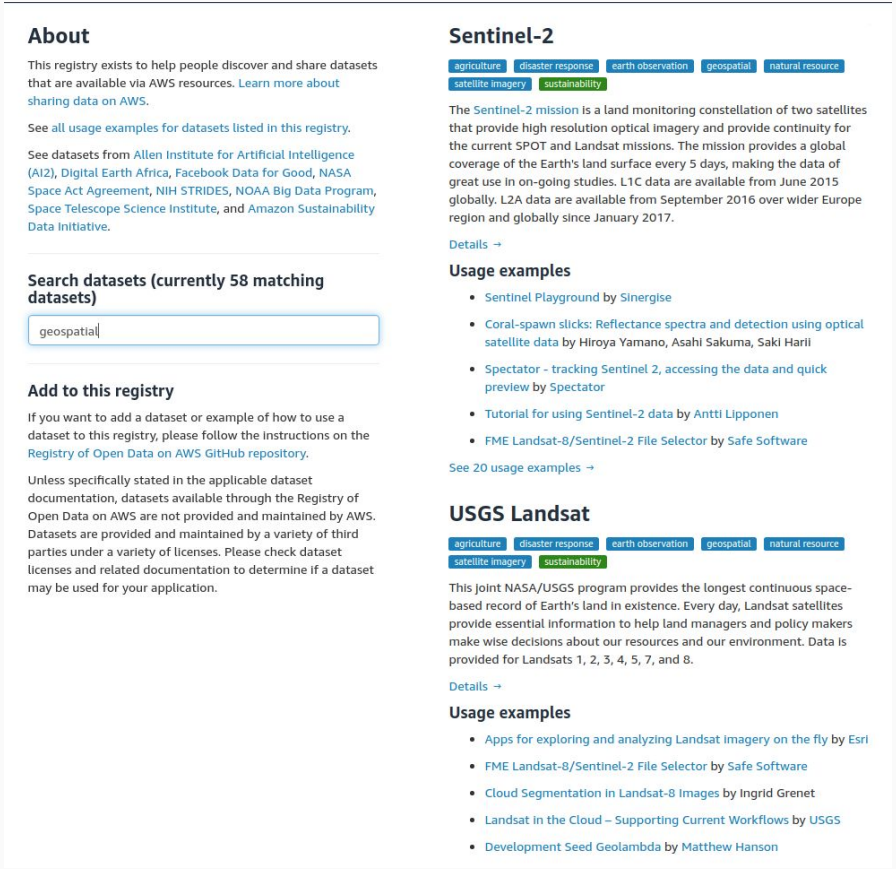
NASA Archive



	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	2025
Archive Size (PB)	15.0	17.7	21.6	26.8	32.0	37.2	55.6	103.4	151.1	198.9	246.6
Annual Growth (PB)	2.6	2.8	3.9	5.2	5.2	5.2	18.4	47.7	47.7	47.7	47.7

If your metadata is not indexed your data does not exist

AWS Open Data



About

This registry exists to help people discover and share datasets that are available via AWS resources. Learn more about sharing data on AWS.

See all usage examples for datasets listed in this registry.

See datasets from Allen Institute for Artificial Intelligence (AI2), Digital Earth Africa, Facebook Data for Good, NASA Space Act Agreement, NIH STRIDES, NOAA Big Data Program, Space Telescope Science Institute, and Amazon Sustainability Data Initiative.

Search datasets (currently 58 matching datasets)

geospatial

Add to this registry

If you want to add a dataset or example of how to use a dataset to this registry, please follow the instructions on the Registry of Open Data on AWS GitHub repository.

Unless specifically stated in the applicable dataset documentation, datasets available through the Registry of Open Data on AWS are not provided and maintained by AWS. Datasets are provided and maintained by a variety of third parties under a variety of licenses. Please check dataset licenses and related documentation to determine if a dataset may be used for your application.

Sentinel-2

agriculture | disaster response | earth observation | geospatial | natural resource | satellite imagery | sustainability

The Sentinel-2 mission is a land monitoring constellation of two satellites that provide high resolution optical imagery and provide continuity for the current SPOT and Landsat missions. The mission provides a global coverage of the Earth's land surface every 5 days, making the data of great use in on-going studies. L1C data are available from June 2015 globally. L2A data are available from September 2016 over wider Europe region and globally since January 2017.

Details →

Usage examples

- Sentinel Playground by Sinergise
- Coral-spawn slicks: Reflectance spectra and detection using optical satellite data by Hiroya Yamano, Asahi Sakuma, Saki Harii
- Spectator - tracking Sentinel 2, accessing the data and quick preview by Spectator
- Tutorial for using Sentinel-2 data by Antti Lipponen
- FME Landsat-8/Sentinel-2 File Selector by Safe Software

See 20 usage examples →

USGS Landsat

agriculture | disaster response | earth observation | geospatial | natural resource | satellite imagery | sustainability

This joint NASA/USGS program provides the longest continuous space-based record of Earth's land in existence. Every day, Landsat satellites provide essential information to help land managers and policy makers make wise decisions about our resources and our environment. Data is provided for Landsats 1, 2, 3, 4, 5, 7, and 8.

Details →

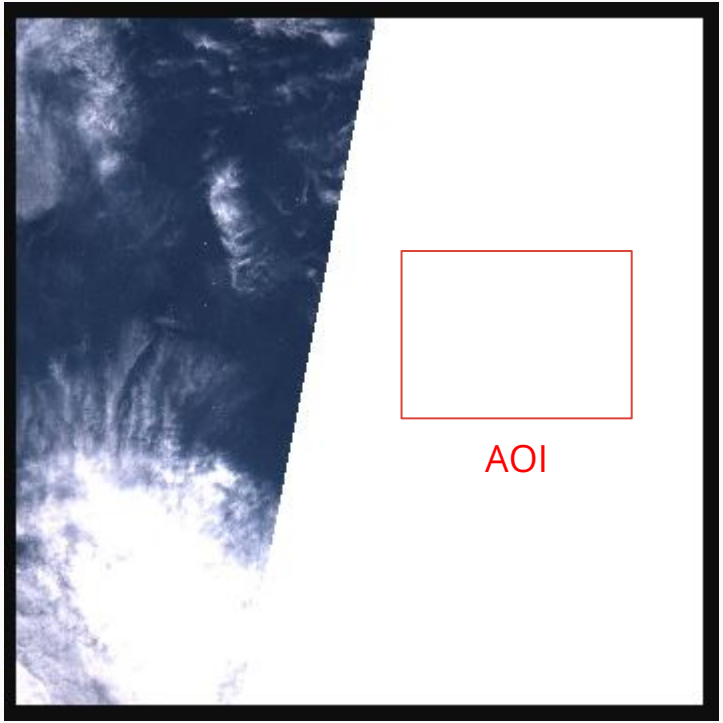
Usage examples

- Apps for exploring and analyzing Landsat imagery on the fly by Esri
- FME Landsat-8/Sentinel-2 File Selector by Safe Software
- Cloud Segmentation in Landsat-8 Images by Ingrid Grenet
- Landsat in the Cloud – Supporting Current Workflows by USGS
- Development Seed Geolambda by Matthew Hanson

Discovery via filename

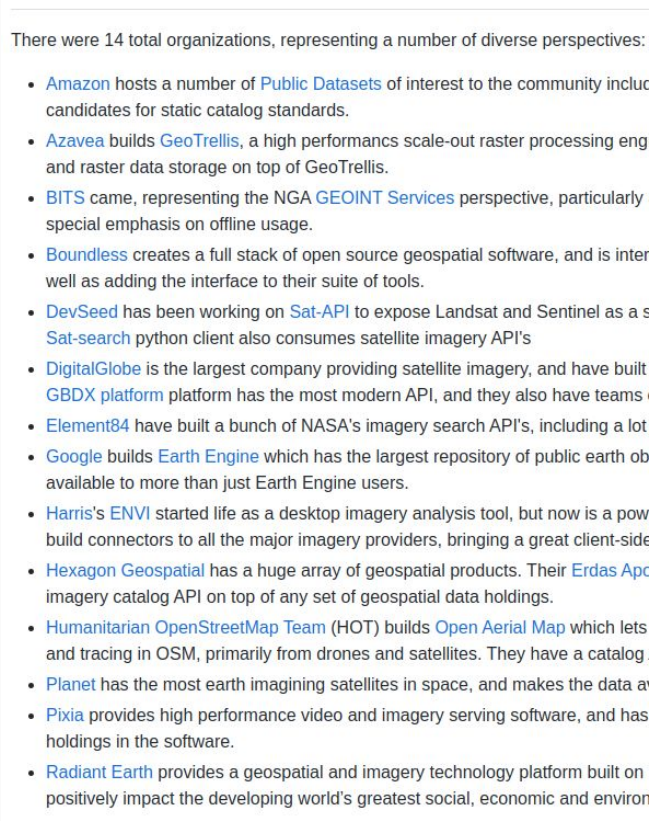
```
matthew@aeon:~$ aws s3 ls s3://sentinel-s2-l2a/tiles/59/M/RT/2020/6/11 --request-payer
 --recursive | more
```

Tile Geometry vs Data Geometry



“Imagery Metadata Spec”

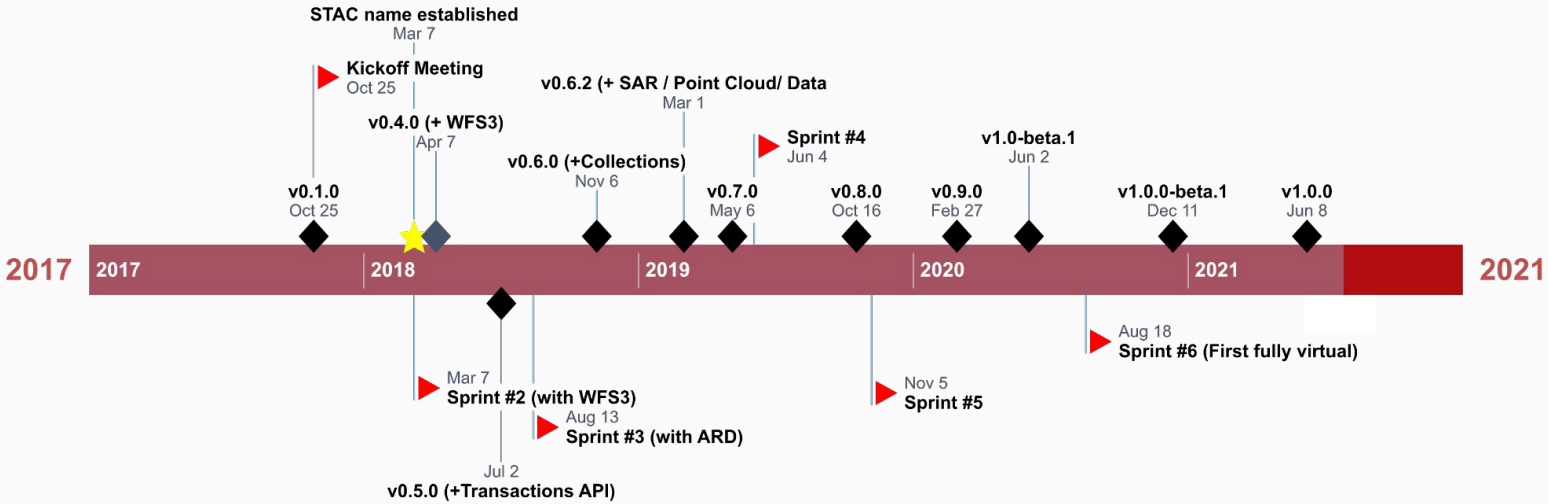
- Boulder, October 2017
- Requirements
  - User-centric search and discovery
  - Leverage other standards (GeoJSON)
  - Not a replacement for existing metadata
  - Crawlable (therefore indexable)
  - Simple extensible core
- Development practices
  - Release early, release often
  - Implement data and software
  - Iterate



There were 14 total organizations, representing a number of diverse perspectives:

- Amazon hosts a number of Public Datasets of interest to the community includ candidates for static catalog standards.
- Azavea builds GeoTrellis, a high performancs scale-out raster processing eng and raster data storage on top of GeoTrellis.
- BITS came, representing the NGA GEOINT Services perspective, particularly special emphasis on offline usage.
- Boundless creates a full stack of open source geospatial software, and is inter well as adding the interface to their suite of tools.
- DevSeed has been working on Sat-API to expose Landsat and Sentinel as a s Sat-search python client also consumes satellite imagery API's
- DigitalGlobe is the largest company providing satellite imagery, and have built GBDX platform platform has the most modern API, and they also have teams
- Element84 have built a bunch of NASA's imagery search API's, including a lot
- Google builds Earth Engine which has the largest repository of public earth ob available to more than just Earth Engine users.
- Harris's ENVI started life as a desktop imagery analysis tool, but now is a pow build connectors to all the major imagery providers, bringing a great client-side
- Hexagon Geospatial has a huge array of geospatial products. Their Erdas Apo imagery catalog API on top of any set of geospatial data holdings.
- Humanitarian OpenStreetMap Team (HOT) builds Open Aerial Map which lets and tracing in OSM, primarily from drones and satellites. They have a catalog
- Planet has the most earth imagining satellites in space, and makes the data a
- Pixia provides high performance video and imagery serving software, and has holdings in the software.
- Radiant Earth provides a geospatial and imagery technology platform built on positively impact the developing world's greatest social, economic and environ

STAC Timeline



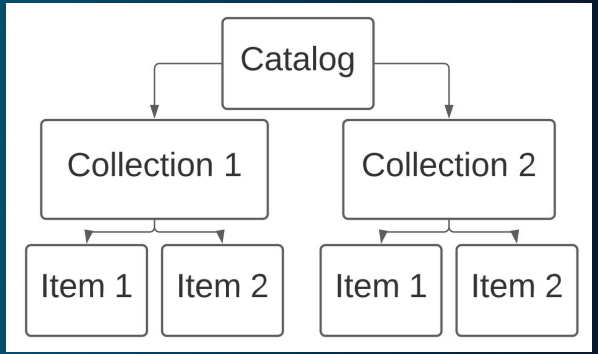
STAC
SpatioTemporal
Asset Catalog

Sponsors


planet.
Radiant Earth Foundation
EARTH IMAGERY FOR IMPACT
Microsoft
geo
Element 84
MAXAR TECHNOLOGIES
ARTURO
USGS
science for a changing world
cosmiQ works
aws
Digital Earth AFRICA
WORLD BANK GROUP
azavea
NASA
UMBRA
open EO
PIXEL8
astræa
NEAR SPACE LABS
THE CLIMATE CORPORATION
VULCAN
Technology
L3HARRIS
Google Earth Engine

SpatioTemporal Asset Catalogs (STAC)

- Specification to describe geospatial data for search, discovery, and use in the cloud
  - JSON
  - Crawlable (indexable)
  - Standardize metadata fields
  - Standard API for querying (OGC API Features extension)



- Catalogs
  - Container that links to other catalogs and collections
  - Top-level catalog is the Root Catalog
- Collections
  - Can contain other catalogs (called sub-catalogs) to group Items further
    - e.g., For Landsat-8 there might be sub-catalog by Landsat Path and Row
  - Contains Items which are similar in some way(s)
    - e.g., same instrument, same region containing data from different instruments, etc.
- Items
  - A single scene or set of datafiles for a specific location at a specific date and time (when/where)

The STAC specifications describe describe geospatial data (stac-spec), and how to searching that data (stac-api-spec)

STAC Status

stac-spec
1.0.0
(May 25, 2021)

OS Projects
30+

stac-api-spec
1.0.0-rc.1
(March 17, 2022)

Extensions
30+

Contributors
50+
(19 w/ 10+ commits)

Public Catalogs
32+
w/ 1+ billion Items

A simple core spec

Field Name	Type	Description
type	string	**REQUIRED.** Type of the GeoJSON Object. MUST be set to `Feature`.
stac_version	string	**REQUIRED.** The STAC version the Item implements.
stac_extensions	[string]	A list of extensions the Item implements.
id	string	**REQUIRED.** Provider identifier. The ID should be unique within the Collection that contains the Item.
geometry	GeoJSON Geometry Object | null	**REQUIRED.** Defines the full footprint of the asset represented by this item, formatted according to RFC 7946, section 3.1. The footprint should be the default GeoJSON geometry, though additional geometries can be included. Coordinates are specified in Longitude/Latitude or Longitude/Latitude/Elevation based on WGS 84.
bbox	[number]	**REQUIRED if `geometry` is not `null`.** Bounding Box of the asset represented by this Item, formatted according to RFC 7946, section 5.
properties	Properties Object	**REQUIRED.** A dictionary of additional metadata for the Item.
links	[Link Object]	**REQUIRED.** List of link objects to resources and related URLs. A link with the `rel` set to `self` is strongly recommended.
assets	Map<string, Asset Object>	**REQUIRED.** Dictionary of asset objects that can be downloaded, each with a unique key.
collection	string	The `id` of the STAC Collection this Item references to (see `collection` relation type). This field is *required* if such a relation type is present and is *not allowed* otherwise. This field provides an easy way for a user to search for any Items that belong in a specified Collection. Must be a non-empty string.

Data provenance

Canonical location of metadata

STAC metadata for source


```
{
   type: "Feature",
   stac_version: "1.0.0-beta.2",
 - stac_extensions: [
       "eo",
       "view",
       "proj"
   ],
   id: "S2B_59WMN_20201102_0_L2A",
 + bbox: […],
 + geometry: {…},
 + properties: {…},
   collection: "sentinel-s2-l2a-cogs",
 + assets: {…},
 - links: [
     - {
         rel: "self",
         href: "https://earth-search.aws.element84.com/v0/collections/sentinel-s2-l2a-cogs/items/S2B_59WMN_20201102_0_L2A"
       },
     - {
         rel: "canonical",
         href: "https://sentinel-cogs.s3.us-west-2.amazonaws.com/sentinel-s2-l2a-cogs/59/W/MN/2020/11/S2B_59WMN_20201102_0_L2A/S2B_59WMN_20201102_0_L2A.json",
         type: "application/json"
       },
     - {
         title: "sentinel-s2-l2a-aws/workflow-publish-sentinel/tiles-59-W-MN-2020-11-2-0",
         rel: "via-cirrus",
         href: "https://cirrus-earth-search.aws.element84.com/v0/catid/sentinel-s2-l2a-aws/workflow-publish-sentinel/tiles-59-W-MN-2020-11-2-0"
       },
     - {
         title: "Source STAC Item",
         rel: "derived_from",
         href: "https://cirrus-v0-data-1qm7qekzjucbq.s3.us-west-2.amazonaws.com/sentinel-s2-l2a/59/W/MN/2020/11/S2B_59WMN_20201102_0_L2A/S2B_59WMN_20201102_0_L2A.json",
         type: "application/json"
       },
     - {
         title: "sentinel-s2-l2a/workflow-cog-archive/S2B_59WMN_20201102_0_L2A",
         rel: "via-cirrus",
         href: "https://cirrus-earth-search.aws.element84.com/v0/catid/sentinel-s2-l2a/workflow-cog-archive/S2B_59WMN_20201102_0_L2A"
       },
     - {
         rel: "parent",
         href: "https://earth-search.aws.element84.com/v0/collections/sentinel-s2-l2a-cogs"
       },
     - {
         rel: "collection",
         href: "https://earth-search.aws.element84.com/v0/collections/sentinel-s2-l2a-cogs"
       },
     - {
         rel: "root",
         href: "https://earth-search.aws.element84.com/v0/"
       }
   ]
}
```

stac-extensions.github.io

STAC Extensions

The stac-extensions github organization is a home for extensions to the SpatioTemporal Asset Catalog specification.

To learn about STAC and Extensions start with the extensions section of the core specification. It explains how extensions work, lists all the known extensions, and has instructions for how to go about 'extending STAC'.

The stac-extensions github organization is a home for many of the leading 'community extensions', providing a neutral home for collaboration. Many of these used to be in the core stac-spec repository, but were removed for 1.0.0 so they could evolve at their own pace, instead of having to follow the core STAC release cycle.

List of STAC Extensions

The definitive list of STAC Extensions is in the core spec repository, in the Extensions README. This includes the official core extensions, as well as a complete list of 'community' extensions. A subset of the community extensions use this stac-extensions GitHub organization, and those are listed in the next section.

Extensions in stac-extensions organization

These extensions add new fields or semantics to STAC objects.

Extension Title	Field Name Prefix	Scope	Maturity	Published?	Description
CARD4L	card4l.	Item	*Proposal*	No	How to comply to the CEOS CARD4L product family specifications (Optical and SAR)
Data Cube	cube	Item, Collection	*Proposal*	Yes	Data Cube related metadata, especially to describe their dimensions

Notable Extensions

projection

Field Name	Type	Description
proj:epsg	integer|null	**REQUIRED.** EPSG code of the datasource
proj:wkt2	string|null	WKT2 string representing the Coordinate Reference System (CRS) that the `proj:geometry` and `proj:bbox` fields represent
proj:projjson	PROJJSON Object|null	PROJJSON object representing the Coordinate Reference System (CRS) that the `proj:geometry` and `proj:bbox` fields represent
proj:geometry	GeoJSON Geometry Object	Defines the footprint of this Item.
proj:bbox	[number]	Bounding box of the Item in the asset CRS in 2 or 3 dimensions.
proj:centroid	Centroid Object	Coordinates representing the centroid of the Item (in lat/long)
proj:shape	[integer]	Number of pixels in Y and X directions for the default grid
proj:transform	[number]	The affine transformation coefficients for the default grid

Electro-Optical

Field Name	Type	Description
eo:bands	[Band Object]	An array of available bands where each object is a Band Object. If given, requires at least one band.
eo:cloud_cover	number	Estimate of cloud cover

SAR

Field Name	Type	Description
sar:instrument_mode	string	**REQUIRED.** The name of the sensor acquisition mode that is commonly used. This should be the short name, if available. For example, `WV` for "Wave mode" of Sentinel-1 and Envisat ASAR satellites.
sar:frequency_band	string	**REQUIRED.** The common name for the frequency band to make it easier to search for bands across instruments. See section "Common Frequency Band Names" for a list of accepted names.
sar:center_frequency	number	The center frequency of the instrument, in gigahertz (GHz).
sar:polarizations	[string]	**REQUIRED.** Any combination of polarizations.
sar:product_type	string	**REQUIRED.** The product type, for example `SSC`, `MGD`, or `SGC`
sar:resolution_range	number	The range resolution, which is the maximum ability to distinguish two adjacent targets perpendicular to the flight path, in meters (m).
sar:resolution_azimuth	number	The azimuth resolution, which is the maximum ability to distinguish two adjacent targets parallel to the flight path, in meters (m).
sar:pixel_spacing_range	number	The range pixel spacing, which is the distance between adjacent pixels perpendicular to the flight path, in meters (m). Strongly RECOMMENDED to be specified for products of type `GRD`.
sar:pixel_spacing_azimuth	number	The azimuth pixel spacing, which is the distance between adjacent pixels parallel to the flight path, in meters (m). Strongly

raster

Field Name	Type	Description
nodata	number|string	Pixel values used to identify pixels that are nodata in the band either by the pixel value as a number or `nan`, `inf` or `-inf` (all strings).
sampling	string	One of `area` or `point`. Indicates whether a pixel value should be assumed to represent a sampling over the region of the pixel or a point sample at the center of the pixel.
data_type	string	The data type of the pixels in the band. One of the data types as described above.
bits_per_sample	number	The actual number of bits used for this band. Normally only present when the number of bits is non-standard for the `datatype`, such as when a 1 bit TIFF is represented as byte
spatial_resolution	number	Average spatial resolution (in meters) of the pixels in the band.
statistics	Statistics Object	Statistics of all the pixels in the band

Open Geospatial Consortium (OGC) Standards

OGC API (New)	OGC Legacy (Old)	What is it for?
Common	OWS Common	Common API elements
Features	Web Feature Service (WFS)	Serving geospatial vector data
Coverages	Web Coverage Services (WCS)	Serving geospatial raster data
Maps	Web Map Service (WMS)	Serving rendered map layers
Tiles	Web Map Tile Service (WMTS)	Serving data as tiles
Processes	Web Processing Service (WPS)	Invoking geospatial processing requests
Records	Catalog Service for the Web (CSW)	Searching metadata

STAC API

STAC API spec
https://github.com/radiantearth/stac-api-spec

OGC API: Features
+
STAC Extensions

OpenAPI Doc: https://stacspec.org/STAC-ext-api.html

OGC API - Features Endpoints

The core OGC API - Features endpoints are shown below, with details provided in an OpenAPI specification document.

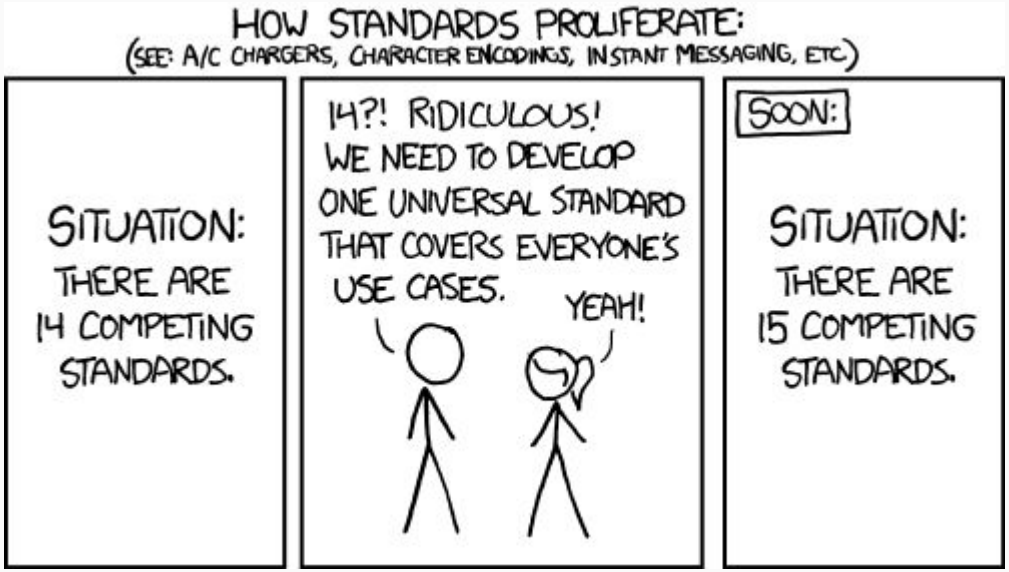
Endpoint	Returns	Description
`/`	JSON	Landing page, links to API capabilities
`/conformance`	JSON	Info about standards to which the API conforms
`/collections`	JSON	Object with a list of Collections contained in the catalog and links
`/collections/{collectionId}`	Collection	Returns single Collection JSON
`/collections/{collectionId}/items`	ItemCollection	GeoJSON FeatureCollection-conformant entity of Items in collection
`/collections/{collectionId}/items/{featureId}`	Item	Returns single Item (GeoJSON Feature)

STAC Endpoints

STAC provides some additional endpoints for the root Catalog itself, as well as the capability to search the Catalog. Note that a STAC API does not need to implement OAFeat, in which case it would only support the endpoints given below. See the OpenAPI specification document.

Endpoint	Returns	Description
`/`	Catalog	Extends `/` from OAFeat to return a full STAC catalog.
`/search`	ItemCollection	Retrieves a group of Items matching the provided search predicates, probably containing search metadata from the `search` extension

Search endpoints


HOW STANDARDS PROLIFERATE:
(SEE: A/C CHARGERS, CHARACTER ENCODINGS, INSTANT MESSAGING, ETC)
SITUATION: THERE ARE 14 COMPETING STANDARDS.
14?! RIDICULOUS! WE NEED TO DEVELOP ONE UNIVERSAL STANDARD THAT COVERS EVERYONE'S USE CASES.
YEAH!
SOON:
SITUATION: THERE ARE 15 COMPETING STANDARDS.

xkcd

Not all standards are equal

Adoption

Users

- NASA (CMR-STAC)
- USGS
- Geoscience Australia
- Microsoft Planetary Computer
- Maxar ARD Delivery
- Arturo
- Astraea
- Capella
- Fastmap
- Google
- Planet
- Hydrosat
- Umbra Space
- Cloud2Street



Chris Holmes @opencholmes · 14m
Replying to @GeoSkeptic
@planet uses @STACspec to distribute some bits of open data, we provide a stactools plugin for python and CLI to convert Planet Orders to STAC, and STAC will likely play a very prominent role in our next generation of API's.

Subit Chakrabarti @abitmore · 4h
Replying to @GeoSkeptic and @opencholmes
At @Cloud2Street we heavily really on STAC for satellite data ingestion and index all the flood maps we produce on a STAC server !
1 4

Scott Soenen @ScottSoenen · 19h
Replying to @GeoSkeptic and @opencholmes
Hydrosat is building on #STAC and plans to make data products available through a STAC API. We're grateful to have shoulders of giants on which to stand.
1 9

Capella Space @capellaspace · 18h
Replying to @GeoSkeptic and @opencholmes
At Capella we use #STAC as a fundamental component of our data access and delivery systems
1 5

Burke Greer @bgr33r · Sep 28
Replying to @GeoSkeptic and @opencholmes
We've got it running in our backend with marblecutter ready for ingest/display at @GeositeInc. This means we're interoperable against any STAC-friendly sources.
2 9

Ben Tuttle
@bentuttle1
Replying to @GeoSkeptic and @opencholmes
STAC is at the core of our geospatial services at @arturo_ai and we will continue to leverage and contribute to it as we develop new capabilities (both internal and customer facing).

Joe Morrison @mouthofmorrison · 21h
Replying to @GeoSkeptic and @opencholmes
We're building our archive on top of STAC and even exploring making our tasking interface as STAC-like as possible. @npcasler and @notthatbreezy would know better than me.
1 5

Rema @nearspacerema · Sep 28
Replying to @GeoSkeptic @HamedAlemo and @opencholmes
Yep, we do @NearSpaceLabs
1 6

James O'Connor @James_O_Connor1 · Sep 28
Replying to @GeoSkeptic and @opencholmes
At @satellitevu we have a stac-fastapi deployment running internally for our flight trial data and are planning a release of a publically accessible static catalog of sample imagery. We're prelaunch still, but absolutely going to use it in the future
1 2 10

Joe Morrison @mouthofmorrison · 21h
nice!
2

Jan Žák @zakjan · 12h
Replying to @GeoSkeptic and @opencholmes
@WeatherLayers client API is based on STAC with additional fields to describe default visualization config (raster, contours, highs/lows, particle animation). Raster part is similar to 'color palette' from the closed 'rendering hints' PR, but there is more

> **Chris Holmes** @opencholmes · Aug 30
> Replying to @zakjan @DanielJDufour and 3 others
> Yes, that's a good place to start. It may end making sense to be its own related extension. I also had a never finished 'rendering hints' extension that could fit this. Most of the fields there ended up making more sense in raster, so not much was left to push forward.

2

Marc Pfister @marcpfister · 20h
Replying to @GeoSkeptic and @opencholmes
At Maxar, STAC files are a core part of our ARD product: ard.maxar.com/docs/about/#im.... They're the peanut butter to the COG jelly.
1 1 9

Alex Leith @alexgleith · 15h
haha, love it!

Although I think I'd prefer the STAC sauce to the COG snag!
1 1

Show replies

Zac Deziel @zacdezgeo · 14h
Replying to @GeoSkeptic and @opencholmes
Fastmap, a Calgary based startup, is building their customer facing API based on STAC
1 4

Jason Suwala @jsuwala · Sep 28
Replying to @GeoSkeptic @Element84 and @opencholmes
Yup - at @HatfieldConsult we're using STAC for what we're doing at geoanalytics.ca and

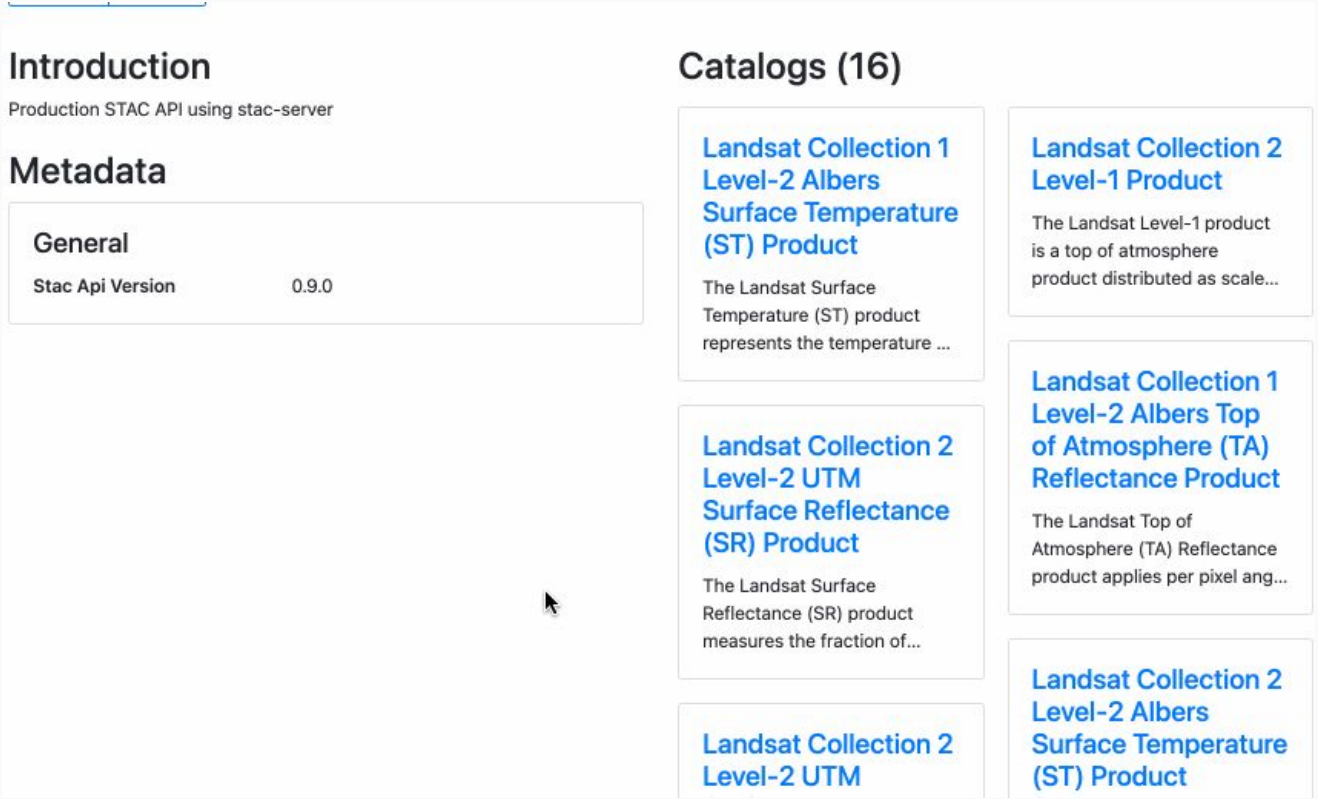
Element 84

Public STAC APIs

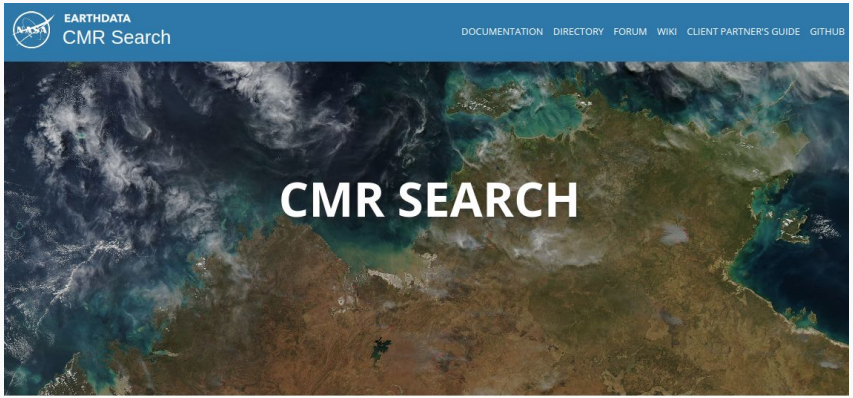
- Earth-Search (Sentinel-2 on AWS)
  - https://earth-search.aws.element84.com/v0
- USGS Landsat (official)
  - https://landsatlook.usgs.gov/stac-server
- NASA Common Metadata Repository (CMR) STAC proxy
  - https://cmr.earthdata.nasa.gov/stac
- Microsoft Planetary Computer
  - https://planetarycomputer.microsoft.com/api/stac/v1
- Google Earth Engine
  - https://earthengine-stac.storage.googleapis.com/catalog/catalog.json
- See STAC Index for more
  - https://stacindex.org/

Landsat Collection 2

https://landsatlook.usgs.gov/stac-server



NASA CMR STAC



The Common Metadata Repository (CMR) is a high-performance, high-quality, continuously evolving metadata system that catalogs Earth Science data and associated service metadata records. These metadata records are registered, modified, discovered, and accessed through programmatic interfaces leveraging standard protocols and APIs.

A Proxy to CMR providing STAC API search and responses


```
{
   id: "stac",
   title: "NASA CMR STAC Proxy",
   stac_version: "1.0.0-beta.1",
   description: "This is the landing page for CMR-STAC. Each provider link below contains a STAC endpoint.",
 - links: [
     - {
         title: "LARC_ASDC",
         rel: "child",
         type: "application/json",
         href: "https://cmr.earthdata.nasa.gov/stac/LARC_ASDC"
       },
     - {
         title: "USGS_EROS",
         rel: "child",
         type: "application/json",
         href: "https://cmr.earthdata.nasa.gov/stac/USGS_EROS"
       },
     - {
         title: "ESA",
         rel: "child",
         type: "application/json",
         href: "https://cmr.earthdata.nasa.gov/stac/ESA"
       },
     - {
         title: "GHRC",
         rel: "child",
         type: "application/json",
         href: "https://cmr.earthdata.nasa.gov/stac/GHRC"
       },
     - {
```

USGS Astrogeology

USGS Controlled Europa Voyager 1, Voyager 2, and Galileo Image Data (usgs_controlled_mosaics_voy1_voy2_galileo)

https://asc-jupiter.s3-us-west-2.amazonaws.com/europa/galileo_voyager/controlled_mosaics/collection.json

The Solid State Imager (SSI) on NASA's Galileo spacecraft acquired more than 500 images of Jupiter's moon, Europa, providing the only moderate- to high-resolution images of the moon's surface. Images were acquired as observation sequences during each orbit that targeted the moon. Each of these observation sequences consists of between 1 and 19 images acquired close in time, that typically overlap, have consistent illumination, and similar pixel scale. The observations vary from relatively low-resolution hemispherical imaging, to high-resolution targeted images that cover a small portion of the surface. Here we provide average mosaics of each of the individual observation sequences acquired by the Galileo spacecraft. These observation mosaics were constructed from a set of 481 Galileo images that were photogrammetrically controlled globally (along with 221 Voyager 1 and 2 images) to improve their relative locations on Europa's surface. The 92 observation mosaics provide users with nearly the entire Galileo Europa imaging dataset at its native resolution and with improved relative image locations.The Solid State Imager (SSI) on NASA's Galileo spacecraft provided the only moderate- to high-resolution images of Jupiter's moon, Europa. Unfortunately, uncertainty in the position and pointing of the spacecraft, as well as the position and orientation of Europa, when the images were acquired resulted in significant errors in image locations on the surface. The result of these errors is that images acquired during different Galileo orbits, or even at different times during the same orbit, are significantly misaligned (errors of up to 100 km on the surface).Previous work has generated global mosaics of Galileo and Voyager images that photogrammetrically control a subset of the available images to correct their relative locations. However, these efforts result in a "static" mosaic that is projected to a consistent pixel scale, and only use a fraction of the dataset (e.g., high resolution images are not included). The purpose of this current dataset is to increase the usability of the entire Galileo image set by photogrammetrically improving the locations of nearly every Europa image acquired by Galileo, and making them available to the community at their native resolution and in easy-to-use regional mosaics based on their acquisition time.The dataset therefore provides a set of image mosaics that can be used for scientific analysis and mission planning activities.



METADATA

STAC Version	1.0.0
License	PDDL-1.0
Temporal Extent	12/16/1997, 4:06:15 AM - 5/22/2000, 4:52:48 PM

ITEM SUMMARY

Instruments	• Galileo Orbiter Solid State Imaging System • Voyager I Soldi State Imager • Voyager II Solid State Imager
Missions	• Galileo Orbiter • Voyager I • Voyager II

Microsoft Planetary Computer

Microsoft | Planetary Computer Explore Data Catalog Hub Applications Documentation Request access

Data Catalog

The Planetary Computer Data Catalog includes petabytes of environmental monitoring data, in consistent, analysis-ready formats. All of the datasets below can be accessed via Azure Blob Storage, and can be used by developers whether you're working within or outside of our Planetary Computer Hub.

Filter by keywords

Datasets available through the Planetary Computer API

Our largest data sets can be queried and accessed through our Planetary Computer API. We are continuing to expand the data available through the API, and continuing to bring new data sets to Azure. If you are interested in seeing additional data on-boarded or published through our API—or if you have data you'd like to contribute—let us know.



Landsat 8 Collection 2 Level-2

Landsat 8 has captured 30m-resolution imagery of the Earth since 2013. This dataset contains global, atmospherically-corrected imagery from Landsat Collection 2.

Landsat USGS NASA Satellite Global Imagery Reflectance



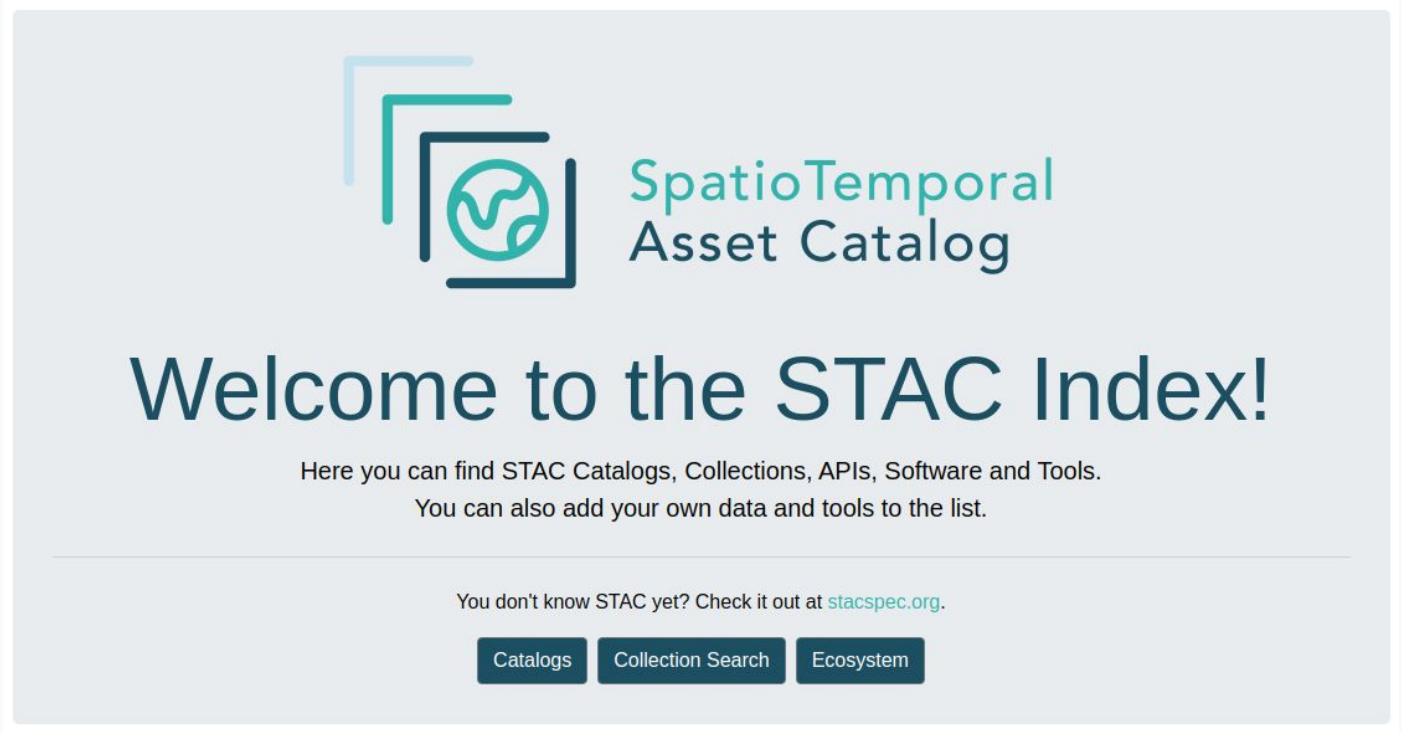
Sentinel-2 Level-2A

The Sentinel-2 program provides global imagery in thirteen spectral bands at 10m-60m resolution and a revisit time of approximately five days. This dataset contains the global Sentinel-2 archive, from 2016 to the present, processed to L2A (bottom-of-atmosphere).

Sentinel Copernicus ESA Satellite Global Imagery Reflectance

Element 84

stacindex.org

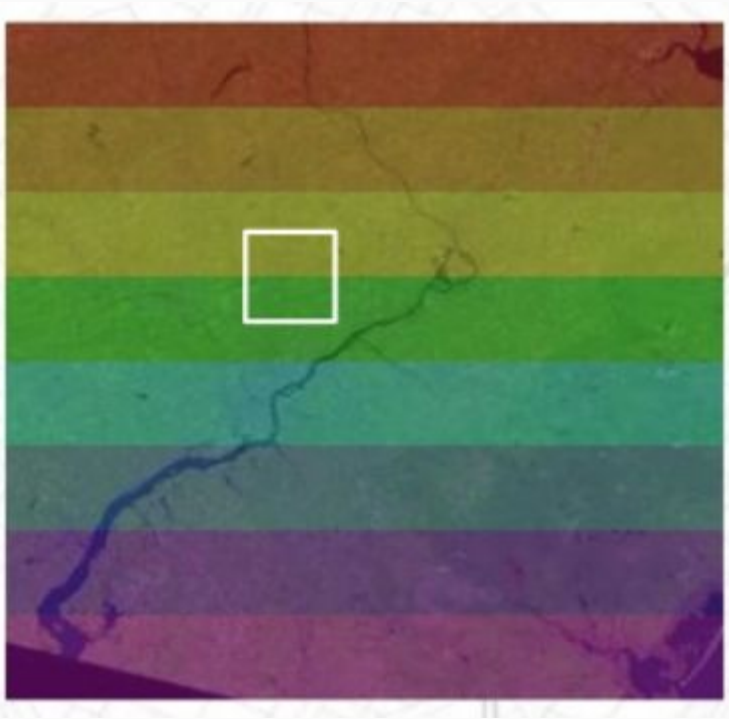


Cloud-Optimized File Formats

Cloud Native Data Formats

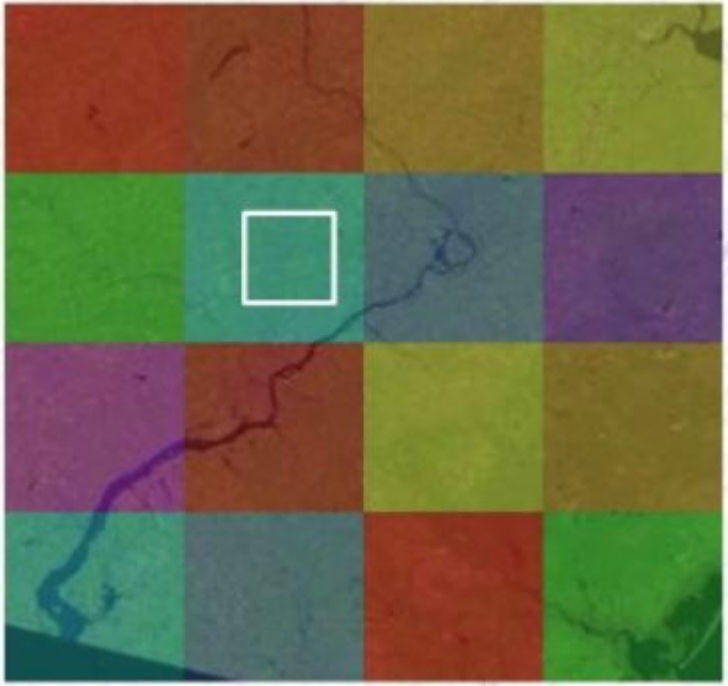
Format	Data Type
Cloud-Optimized GeoTIFF (COG)	Raster (good for EO data)
Zarr	Raster (good for modeled data)
Cloud-Optimized Point Cloud (COPC)	Point Clouds
GeoParquet	Vector
TileDB	Raster, Point Clouds

Striped Chunks





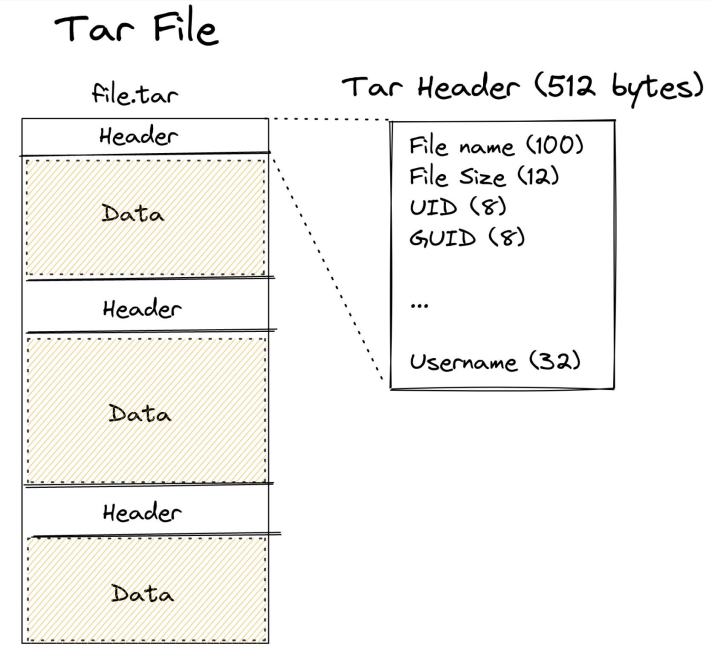
Tiled Chunks

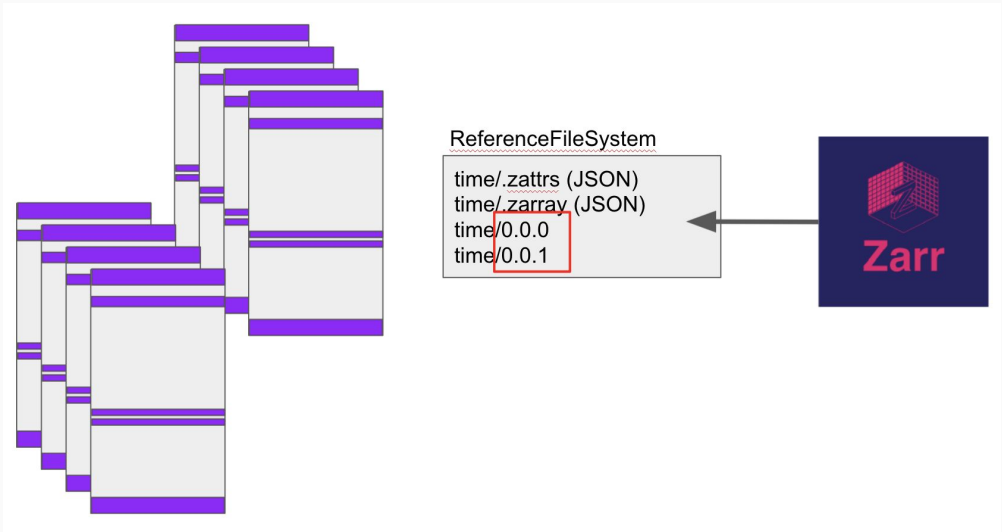




A Cloud Native Dataset is one with small addressable chunks (via files, internal tiles, or both)

Cloud-Optimized <insert-legacy-format>?

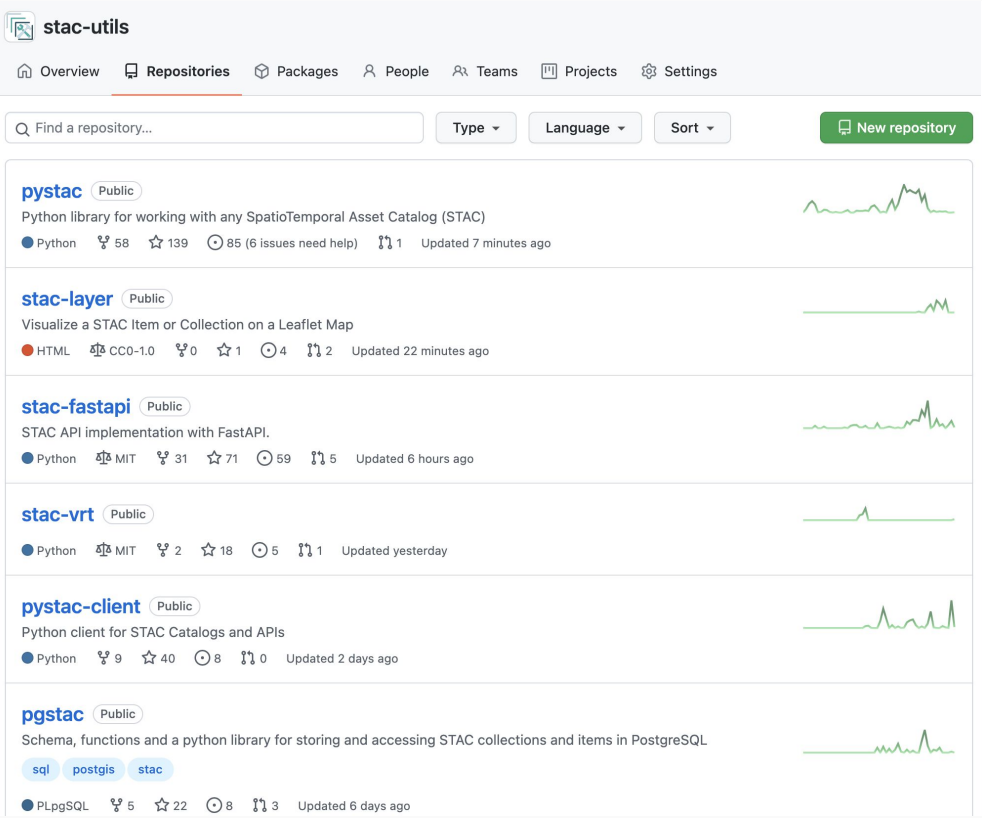


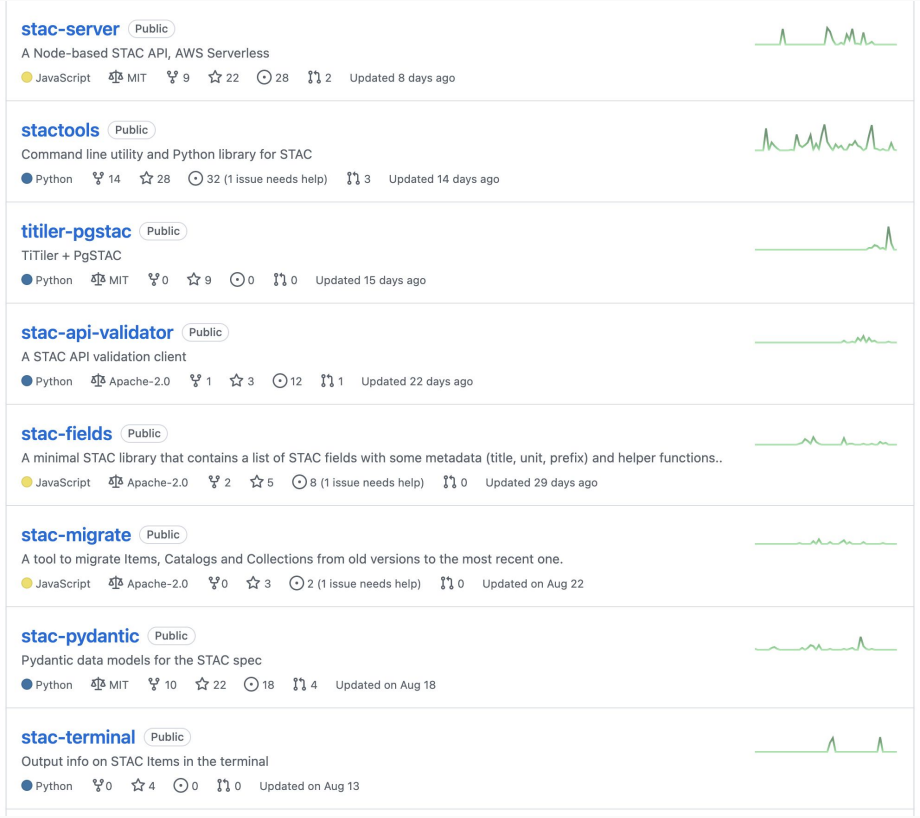


https://github.com/linz/cotar

Open-Source Ecosystem

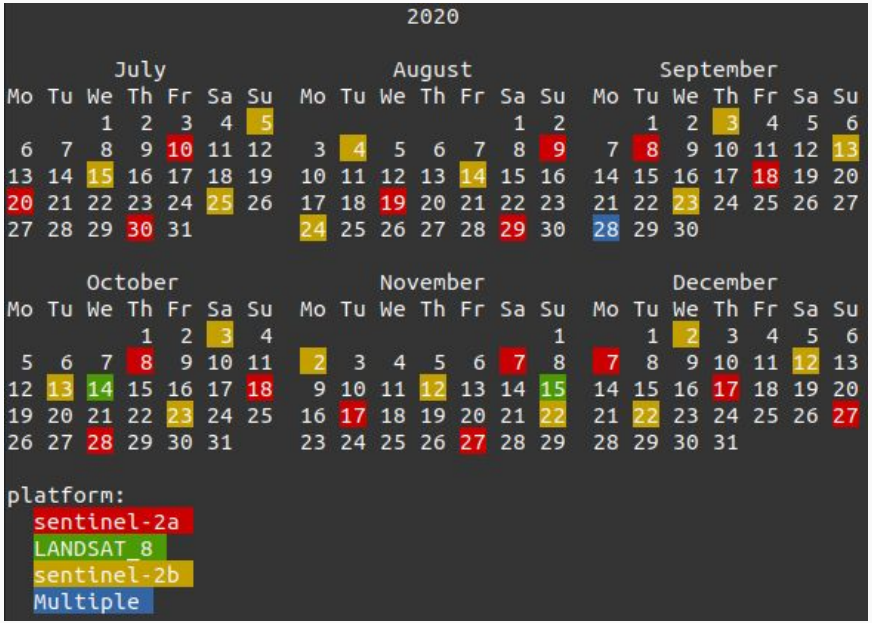
stac-utils


stac-utils
Overview
Repositories
Packages
People
Teams
Projects
Settings
Find a repository...
Type
Language
Sort
New repository
pystac Public
Python library for working with any SpatioTemporal Asset Catalog (STAC)
Python 58 139 85 (6 issues need help) 1 Updated 7 minutes ago
stac-layer Public
Visualize a STAC Item or Collection on a Leaflet Map
HTML CC0-1.0 0 1 4 2 Updated 22 minutes ago
stac-fastapi Public
STAC API implementation with FastAPI.
Python MIT 31 71 59 5 Updated 6 hours ago
stac-vrt Public
Python MIT 2 18 5 1 Updated yesterday
pystac-client Public
Python client for STAC Catalogs and APIs
Python 9 40 8 0 Updated 2 days ago
pgstac Public
Schema, functions and a python library for storing and accessing STAC collections and items in PostgreSQL
sql postgis stac
PLpgSQL 5 22 8 3 Updated 6 days ago


stac-server Public
A Node-based STAC API, AWS Serverless
JavaScript MIT 9 22 28 2 Updated 8 days ago
stactools Public
Command line utility and Python library for STAC
Python 14 28 32 (1 issue needs help) 3 Updated 14 days ago
titiler-pgstac Public
TiTiler + PgSTAC
Python MIT 0 9 0 0 Updated 15 days ago
stac-api-validator Public
A STAC API validation client
Python Apache-2.0 1 3 12 1 Updated 22 days ago
stac-fields Public
A minimal STAC library that contains a list of STAC fields with some metadata (title, unit, prefix) and helper functions..
JavaScript Apache-2.0 2 5 8 (1 issue needs help) 0 Updated 29 days ago
stac-migrate Public
A tool to migrate Items, Catalogs and Collections from old versions to the most recent one.
JavaScript Apache-2.0 0 3 2 (1 issue needs help) 0 Updated on Aug 22
stac-pydantic Public
Pydantic data models for the STAC spec
Python MIT 10 22 18 4 Updated on Aug 18
stac-terminal Public
Output info on STAC Items in the terminal
Python 0 4 0 0 Updated on Aug 13

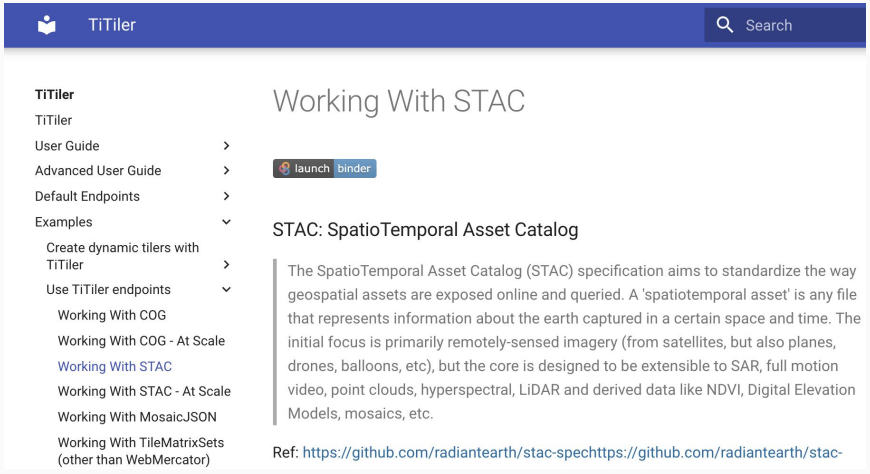
pystac-client

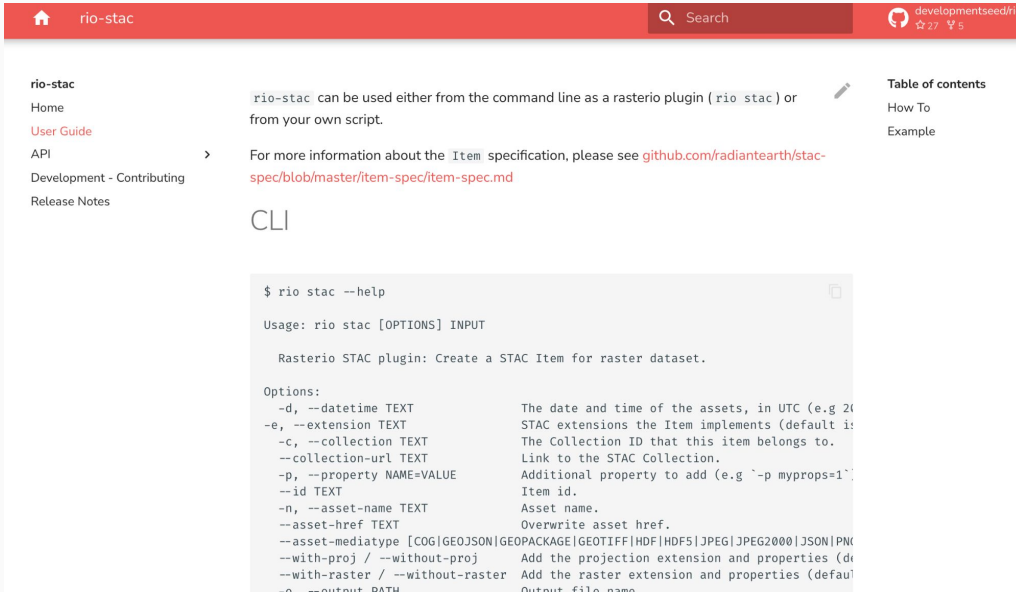
```
$ export STAC_API_URL=https://earth-search.aws.element84.com/v0
$ stac-client search --intersects aoi.json --datetime 2020-07-01/2020-12-31 -c sentinel-s2-l2a-cogs landsat-8-l1-c1 | stacterm cal --label platform
```



TiTiler and Rasterio

- Rasterio's 'rio stac' will create a basic STAC Item with all the key raster information
- Titiler exposes STAC endpoints, and also can work with STAC data
- Enables advanced workflows when combined with stac-fastapi to render mosaics of STAC API search results on the fly





Element 84

stactools

High level CLI/lib for generating and working with STAC

To install all available sub-packages, use

```
pip install stactools[all]
```

`stactools` Features ¶

Use `stac --help` to navigate the varios commands.

- `stac copy` copies STACs and optionally assets
- `stac info` and `stac describe` display information about STACs
- `stac layout` will modify the layout of a STAC based on item properties

One of the focuses of `stactools` is to provide an easy way to plug in functionality for different data sources. For example:

- `stac planet convert-order` will convert a Planet order into a STAC.

See the Command Line Interface documentation for more details about each command.

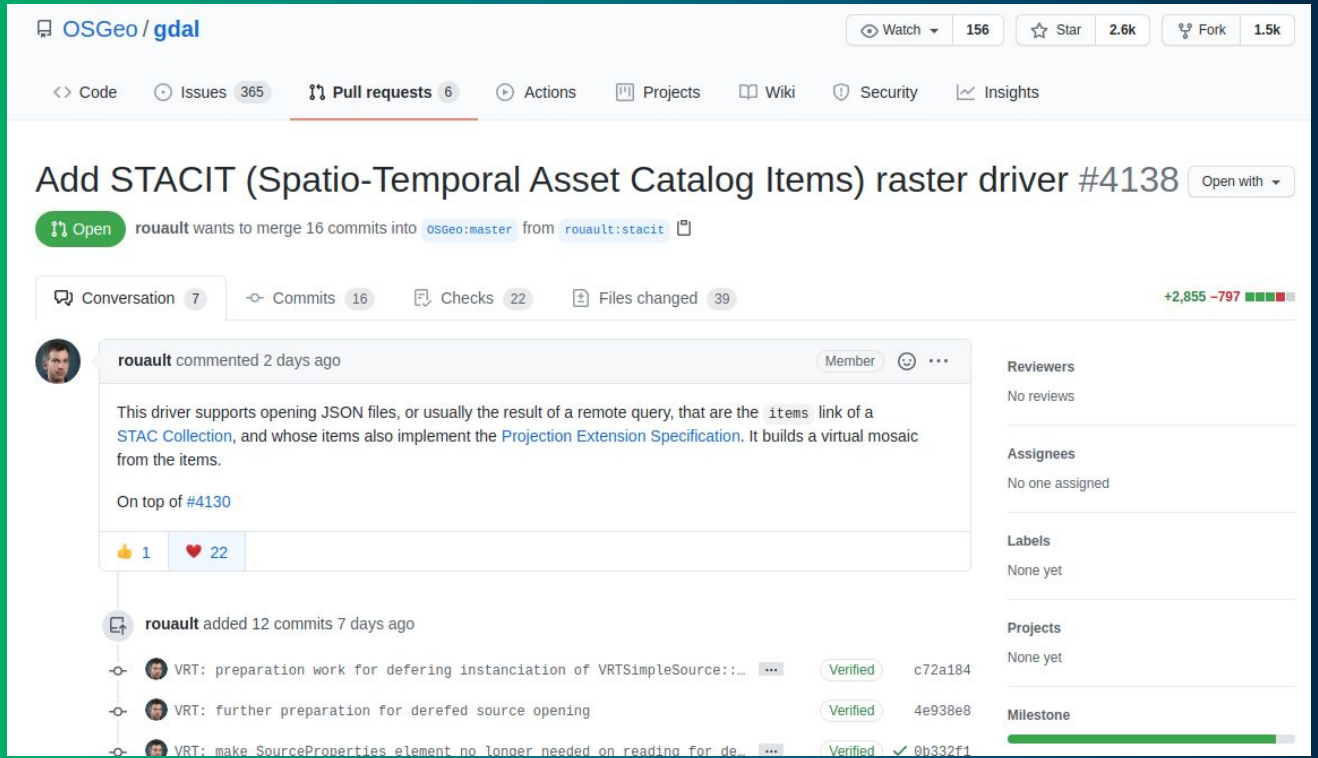
Table of Contents

- API Reference
  - Core

`stactools` is comprised of subpackages that provide library and CLI functionality. Below is a list of available subpackages.

subpackage	description
`stactools_core`	Contains core library functionality that is used across the other projects
`stactools_cli`	Contains the command line interface (cli) for running the `stactools` command
`stactools_aster`	Methods and commands for working with ASTER data
`stactools_corine`	Methods and commands for working with CORINE Land Cover data
`stactools_planet`	Methods and commands for working with planet data
`stactools_landsat`	Methods and commands for working with landsat data
`stactools_browse`	Contains a command for launching stac-browser against a local STAC

GDAL Driver!


OSGeo / gdal
Watch 156
Star 2.6k
Fork 1.5k
Code
Issues 365
Pull requests 6
Actions
Projects
Wiki
Security
Insights
Add STACIT (Spatio-Temporal Asset Catalog Items) raster driver #4138
Open with
Open
rouault wants to merge 16 commits into OSGeo:master from rouault:stacit
Conversation 7
Commits 16
Checks 22
Files changed 39
+2,855 −797
rouault commented 2 days ago
Member
This driver supports opening JSON files, or usually the result of a remote query, that are the items link of a STAC Collection, and whose items also implement the Projection Extension Specification. It builds a virtual mosaic from the items.
On top of #4130
1
22
rouault added 12 commits 7 days ago
VRT: preparation work for defering instanciation of VRTSimpleSource::…
Verified
c72a184
VRT: further preparation for derefed source opening
Verified
4e938e8
VRT: make SourceProperties element no longer needed on reading for de…
Verified
0b332f1
Reviewers
No reviews
Assignees
No one assigned
Labels
None yet
Projects
None yet
Milestone

STAC Browser v3

Planet Disaster Data

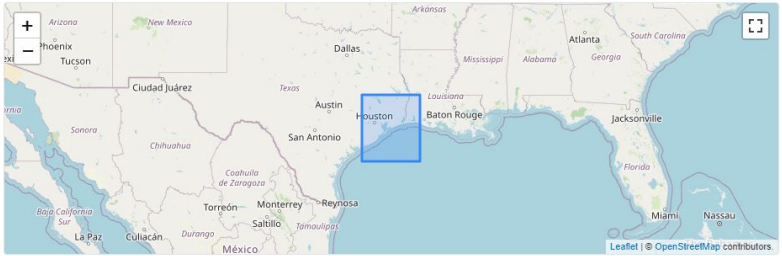
Browse

Introduction

Planet Disaster Data makes imagery available directly to the public, volunteers, humanitarian organizations, and other coordinating bodies in support of the International Charter for Space and Major Disasters. Data is released for individual disaster events, providing a 30 day window pre- and post-disaster. Imagery is provided under Creative Commons licenses, free of charge, with either CC-BY-SA or CC-BY-NC.

disaster open

Licenses	CC-BY-SA-4.0
Temporal Extents	28.8.2017, 18:00:00 UTC until present



Providers

- Planet PRODUCER PROCESSOR
- Planet Disaster Team <disaster-team@planet.com> LICENSOR
- Google Cloud Platform HOST

Catalogs (1)

Hurricane Harvey

Planet Disaster Data makes imagery available directly to the public, volunteers, humanitarian...

Metadata

General

Constellation	1. planetscope 2. skysat
Platform	1. 101c 2. SS02 3. SSC1

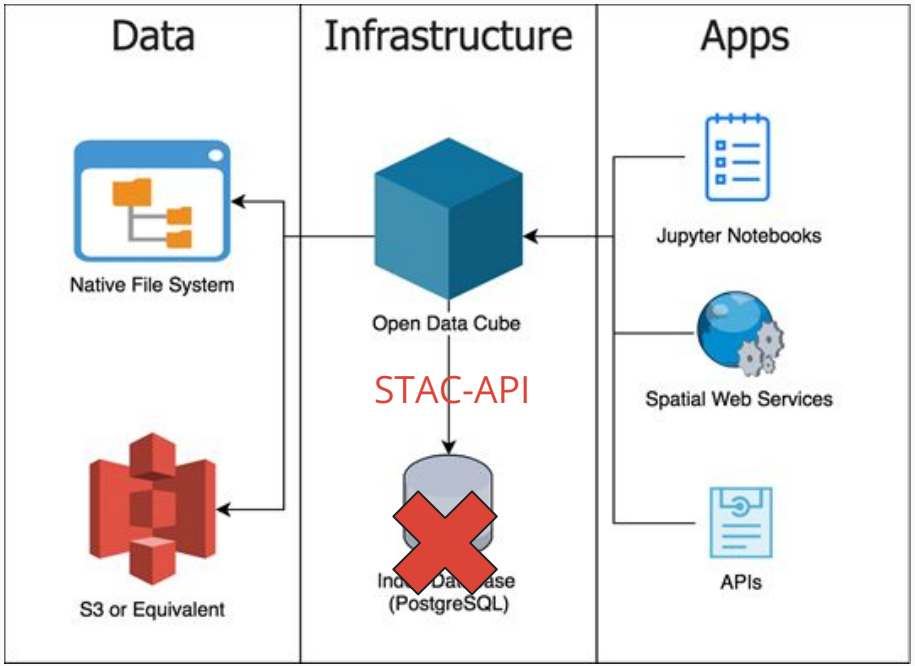
Electro-Optical

Gsd	0,9 – 3,7
Cloud Cover	0 % – 24 %

View Geometry

Off-Nadir Angle	0,2 ° – 27,3 °
Sun Elevation	56,3 ° – 65,1 °
Sun Azimuth	122 ° – 231,9 °

ODC-STAC (OpenDataCube)



Element 84

Australian Government
Geoscience Australia

Digital Earth
AFRICA

Find STAC Items to Load

[3]:
```
km2deg = 1.0 / 111
x, y = (113.887, -25.843)  # Center point of a query
r = 100 * km2deg

query = catalog.search(
    collections=["sentinel-s2-l2a-cogs"],
    datetime="2021-09-16",
    limit=10,
    bbox=(x - r, y - r, x + r, y + r),
)

items = list(query.get_items())
print(f"Found: {len(items):d} datasets")
```

```
Found: 9 datasets
```

Construct Dask Dataset

Note that even though there are 9 STAC Items on input, there is only one timeslice on output. This is because of `groupy="solar_day"`. With that setting `stac_load` will place all items that occured on the same day (as adjusted for the timezone) into one image plane.

[4]:
```
# crs = "epsg:32749"  # native UTM projection in the query region
crs = "epsg:3857"  # Since we will plot it on a map we need to use 3857 projection
zoom = 2 ** 5  # overview level 5

xx = stac_load(
    items,
    output_crs=crs,
    resolution=(-10 * zoom, 10 * zoom),
    chunks={},
    groupby="solar_day",
    measurements=["red", "green", "blue"],
    stac_cfg=cfg,
)
xx
```

[4]:

xarray.Dataset

▸ Dimensions: (**time**: 1, **y**: 1098, **x**: 833)

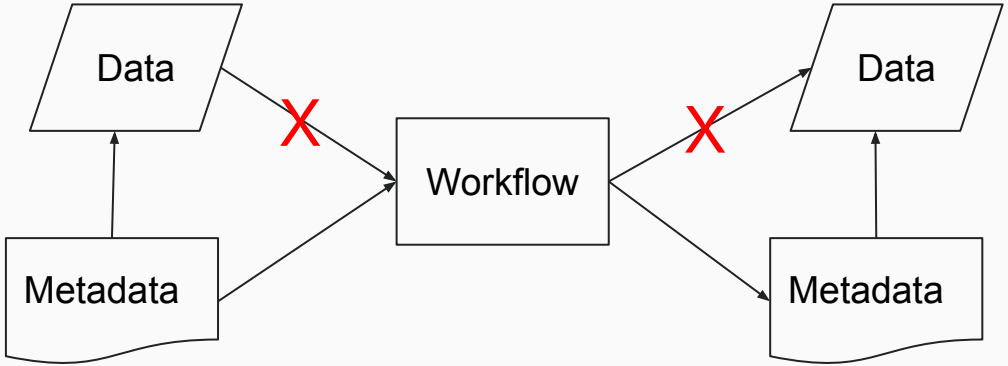
Cloud-Native Geospatial Workflows

Stop thinking of files

```
LC08_L1GT_168005_20190828_20190828_01_RT.tif
LC08_L1GT_168005_20190828_20190828_01_RT_B1.TIF
LC08_L1GT_168005_20190828_20190828_01_RT_B3.TIF
LC08_L1GT_168005_20190828_20190828_01_RT_c10_f1.tif
LC08_L1GT_168005_20190828_20190828_01_RT_cm.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_class_c50_i9.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_tmp_test5.tif
LC08_L1GT_168005_20190828_20190828_01_RT_k1_ver7.6.tif
LC08_L1GT_168005_20190828_20190828_01_final_class2.tif
LC08_L1GT_168005_20190828_20190828_01_final_version_class.tif
LC08_L1GT_168005_20190828_20190828_01_final_version_final.tif
LC08_L1GT_168005_20190828_20190828_01_test.tif
LC08_L1GT_168005_20190828_20190828_test1.tif
LC08_L1GT_168005_20190828_20190828_test_test.tif
```

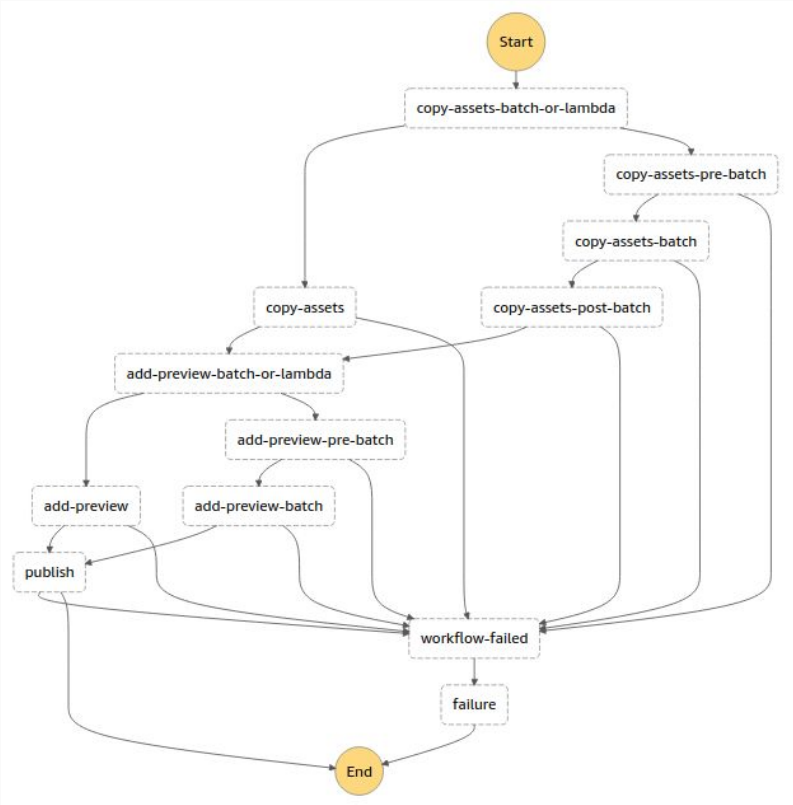
Cloud-Native Geospatial Workflows

- Workflows operate on metadata
  - Metadata in, metadata out
- Metadata contains complete characterization of data
  - Data location and access



STAC Based Pipelines

- Create and publish STAC
  - STAC In, STAC Out
- Maintain data provenance
  - Published STAC Items should point back to source STAC Items
- Chain together workflows
  - Output STAC should be consumable into another workflow



Data provenance

Canonical location of metadata

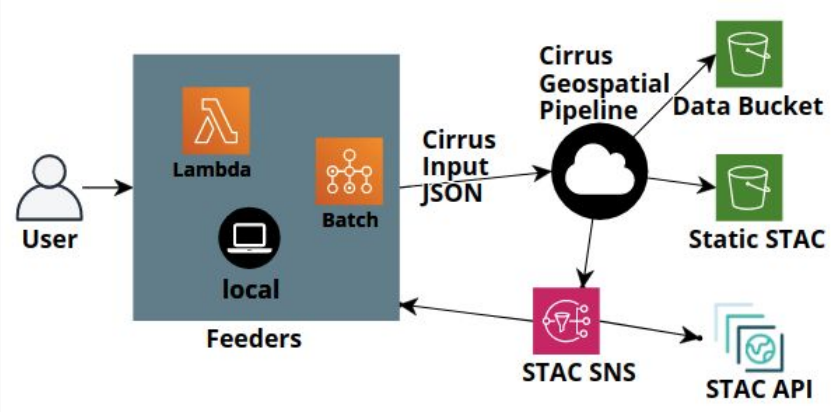
STAC metadata for source

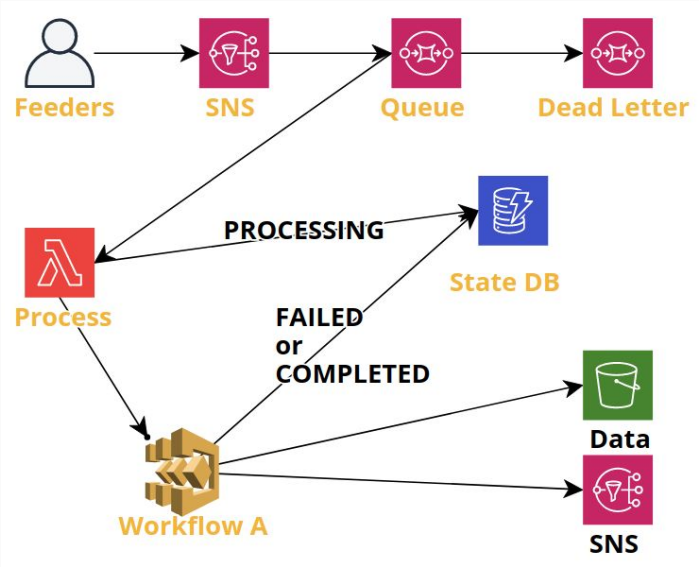
Link to Cirrus record used


```
{
   type: "Feature",
   stac_version: "1.0.0-beta.2",
 - stac_extensions: [
       "eo",
       "view",
       "proj"
   ],
   id: "S2B_59WMN_20201102_0_L2A",
 + bbox: […],
 + geometry: {…},
 + properties: {…},
   collection: "sentinel-s2-l2a-cogs",
 + assets: {…},
 - links: [
     - {
         rel: "self",
         href: "https://earth-search.aws.element84.com/v0/collections/sentinel-s2-l2a-cogs/items/S2B_59WMN_20201102_0_L2A"
       },
     - {
         rel: "canonical",
         href: "https://sentinel-cogs.s3.us-west-2.amazonaws.com/sentinel-s2-l2a-cogs/59/W/MN/2020/11/S2B_59WMN_20201102_0_L2A/S2B_59WMN_20201102_0_L2A.json",
         type: "application/json"
       },
     - {
         title: "sentinel-s2-l2a-aws/workflow-publish-sentinel/tiles-59-W-MN-2020-11-2-0",
         rel: "via-cirrus",
         href: "https://cirrus-earth-search.aws.element84.com/v0/catid/sentinel-s2-l2a-aws/workflow-publish-sentinel/tiles-59-W-MN-2020-11-2-0"
       },
     - {
         title: "Source STAC Item",
         rel: "derived_from",
         href: "https://cirrus-v0-data-1qm7qekzjucbq.s3.us-west-2.amazonaws.com/sentinel-s2-l2a/59/W/MN/2020/11/S2B_59WMN_20201102_0_L2A/S2B_59WMN_20201102_0_L2A.json",
         type: "application/json"
       },
     - {
         title: "sentinel-s2-l2a/workflow-cog-archive/S2B_59WMN_20201102_0_L2A",
         rel: "via-cirrus",
         href: "https://cirrus-earth-search.aws.element84.com/v0/catid/sentinel-s2-l2a/workflow-cog-archive/S2B_59WMN_20201102_0_L2A"
       },
     - {
         rel: "parent",
         href: "https://earth-search.aws.element84.com/v0/collections/sentinel-s2-l2a-cogs"
       },
     - {
         rel: "collection",
         href: "https://earth-search.aws.element84.com/v0/collections/sentinel-s2-l2a-cogs"
       },
     - {
         rel: "root",
         href: "https://earth-search.aws.element84.com/v0/"
       }
   ]
}
```

Cirrus

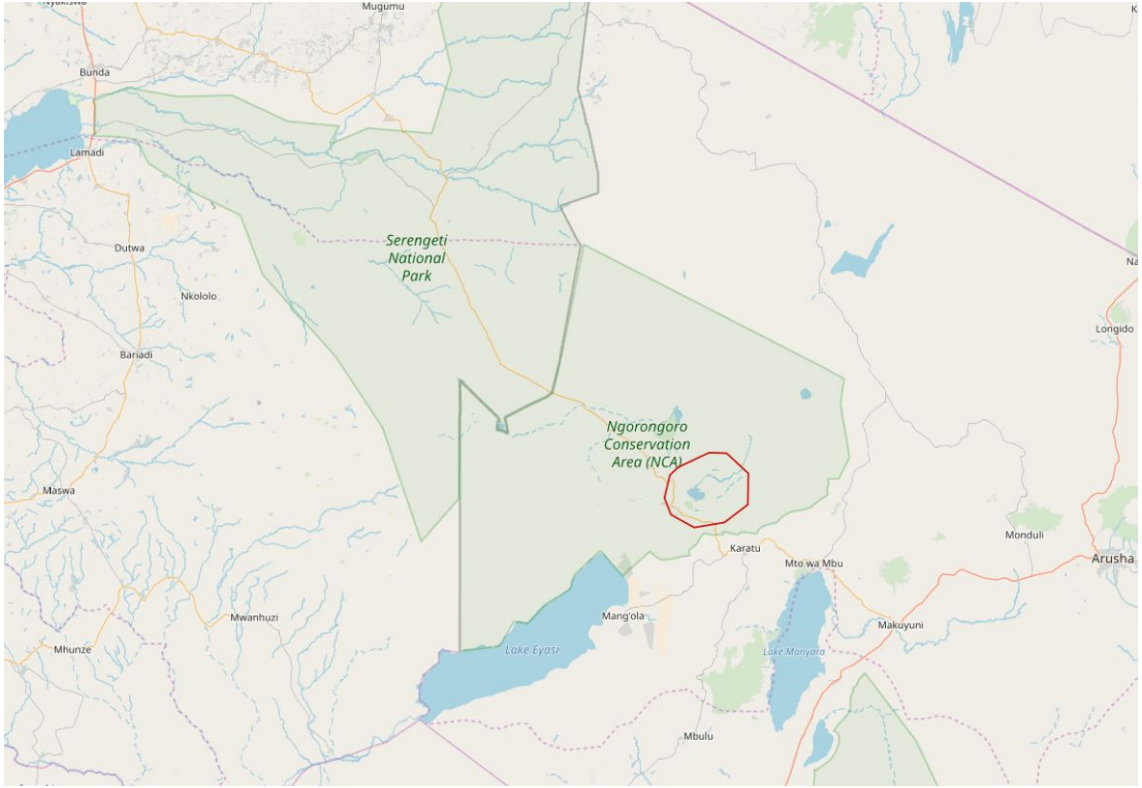
- Open-Source
- Python
- AWS Serverless





Examples

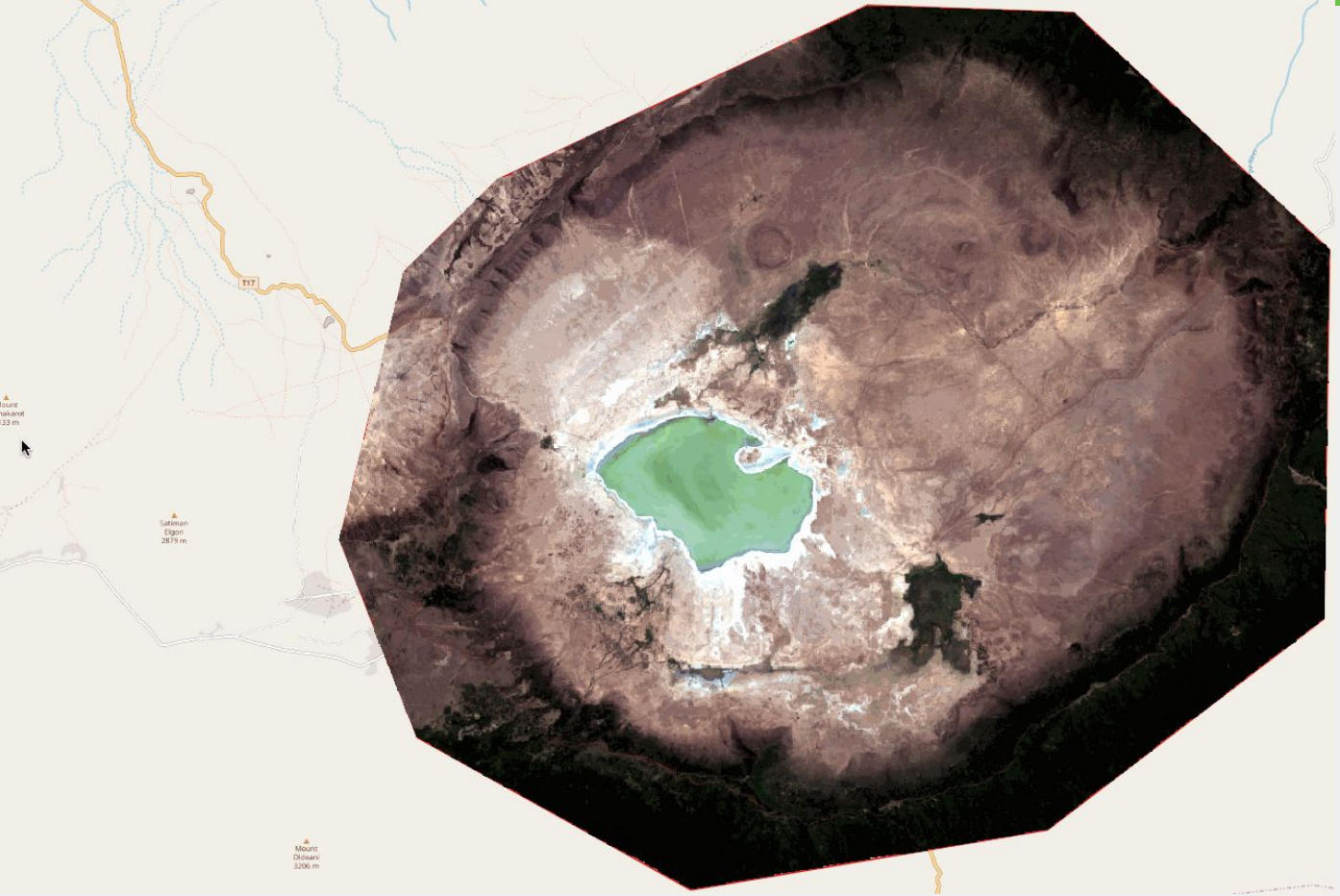
Ngorongoro crater: landsat-8 and sentinel-2



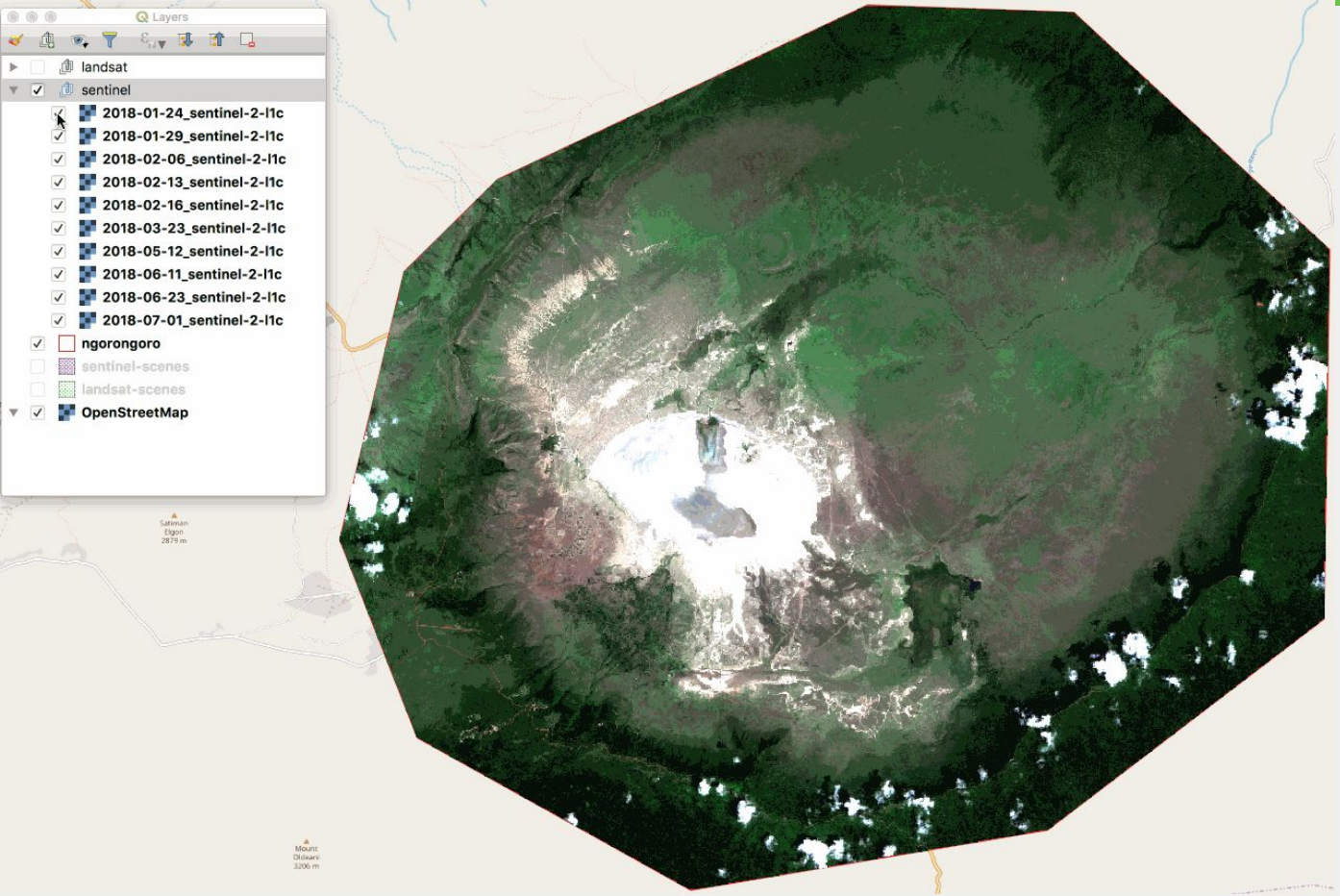
Ngorongoro crater: landsat-8 scenes

```
sat-search search
    \ --intersects ngorongoro.geojson --eo:cloud_cover 0/5 \
    -c landsat-8-l1 --save landsat-scenes.geojson \
    --print_md date eo:cloud_cover id
```

```
2014-11-25  1                LC08_L1TP_169062_20141125_20170417_01
2014-12-11  5                LC08_L1TP_169062_20141211_20170416_01
2015-01-28  1                LC08_L1TP_169062_20150128_20170413_01
2015-02-13  0                LC08_L1TP_169062_20150213_20170413_01
2015-08-24  2                LC08_L1TP_169062_20150824_20170405_01
2015-09-09  1                LC08_L1TP_169062_20150909_20170404_01
2015-12-30  0                LC08_L1TP_169062_20151230_20170331_01
2016-02-16  0                LC08_L1TP_169062_20160216_20180526_01
2016-08-10  2                LC08_L1TP_169062_20160810_20170322_01
2016-10-13  0                LC08_L1TP_169062_20161013_20180526_01
2016-10-29  3                LC08_L1TP_169062_20161029_20170319_01
2017-01-17  0                LC08_L1TP_169062_20170117_20170311_01
2017-11-17  2                LC08_L1TP_169062_20171117_20171122_01
2018-02-05  0                LC08_L1TP_169062_20180205_20180221_01
2018-05-28  3                LC08_L1TP_169062_20180528_20180605_01
18 scenes found
```

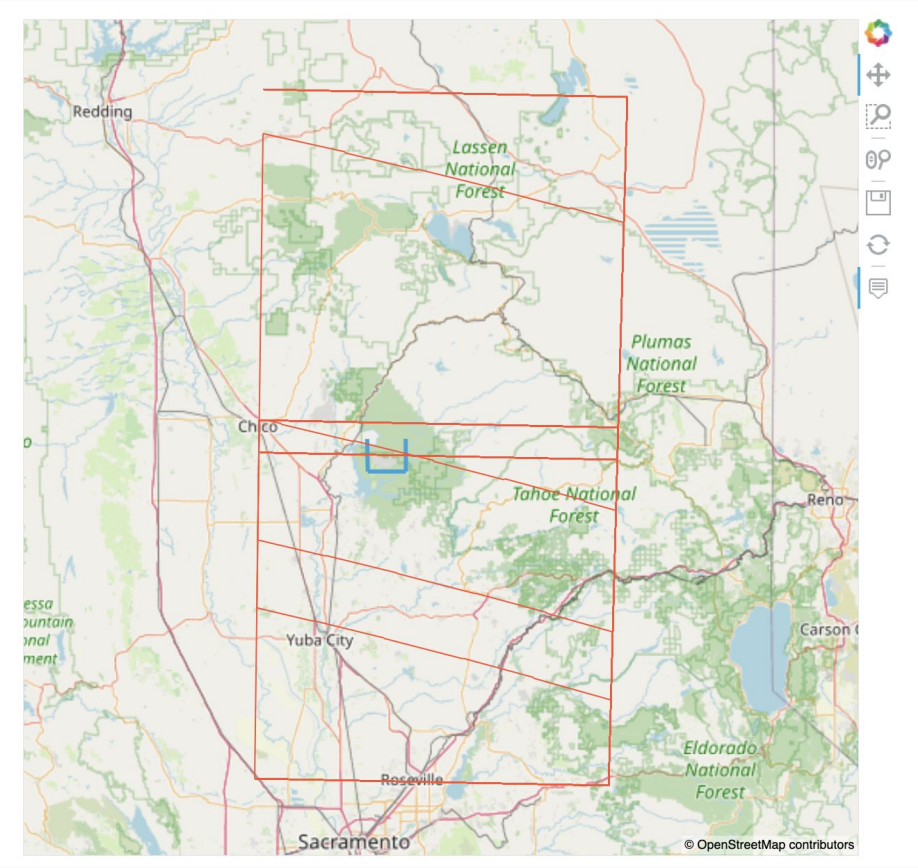


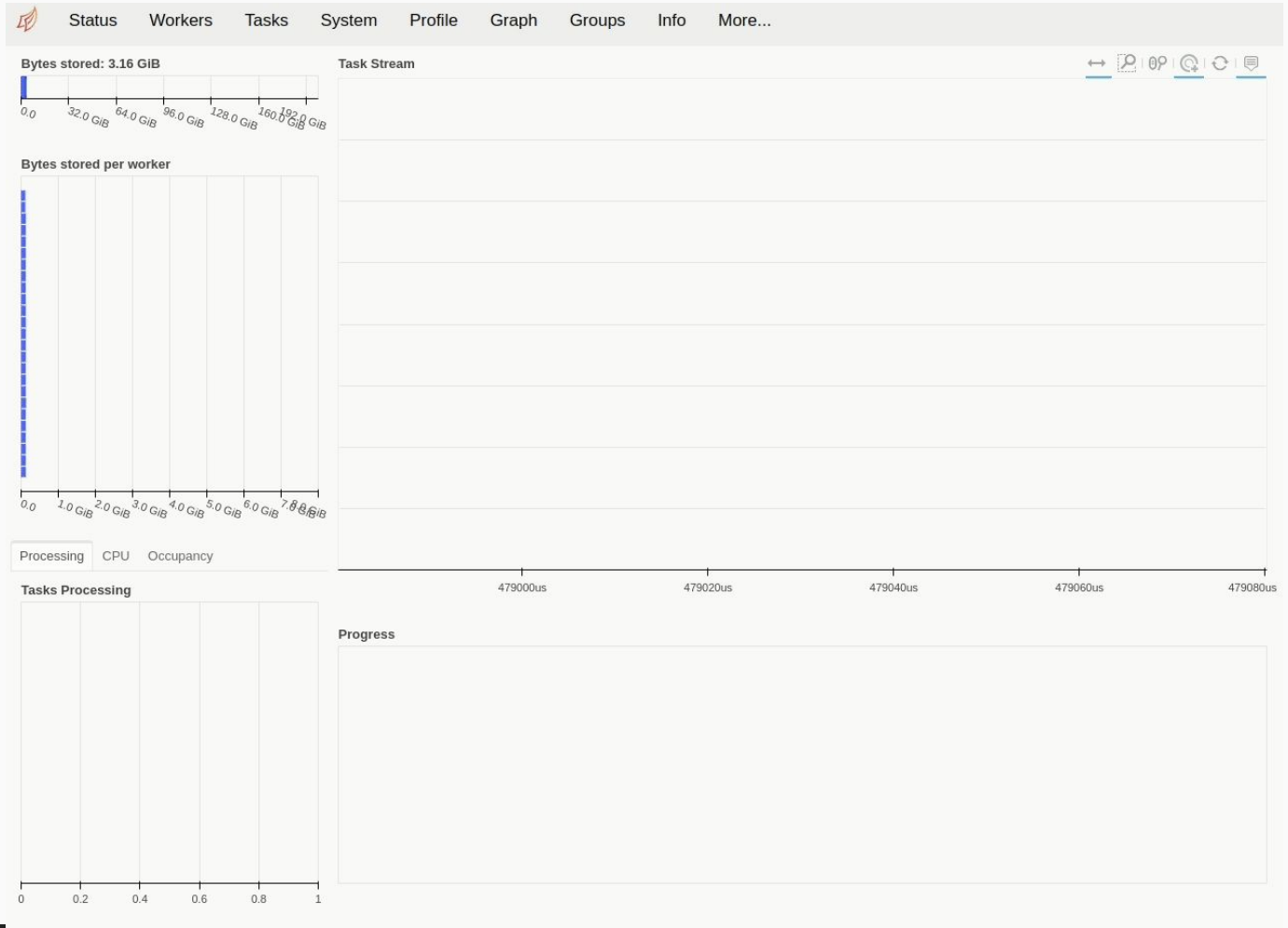

```
$ time sat-fetch load landsat-scenes.geojson \
    --download red green blue
real 0m59.674s
```

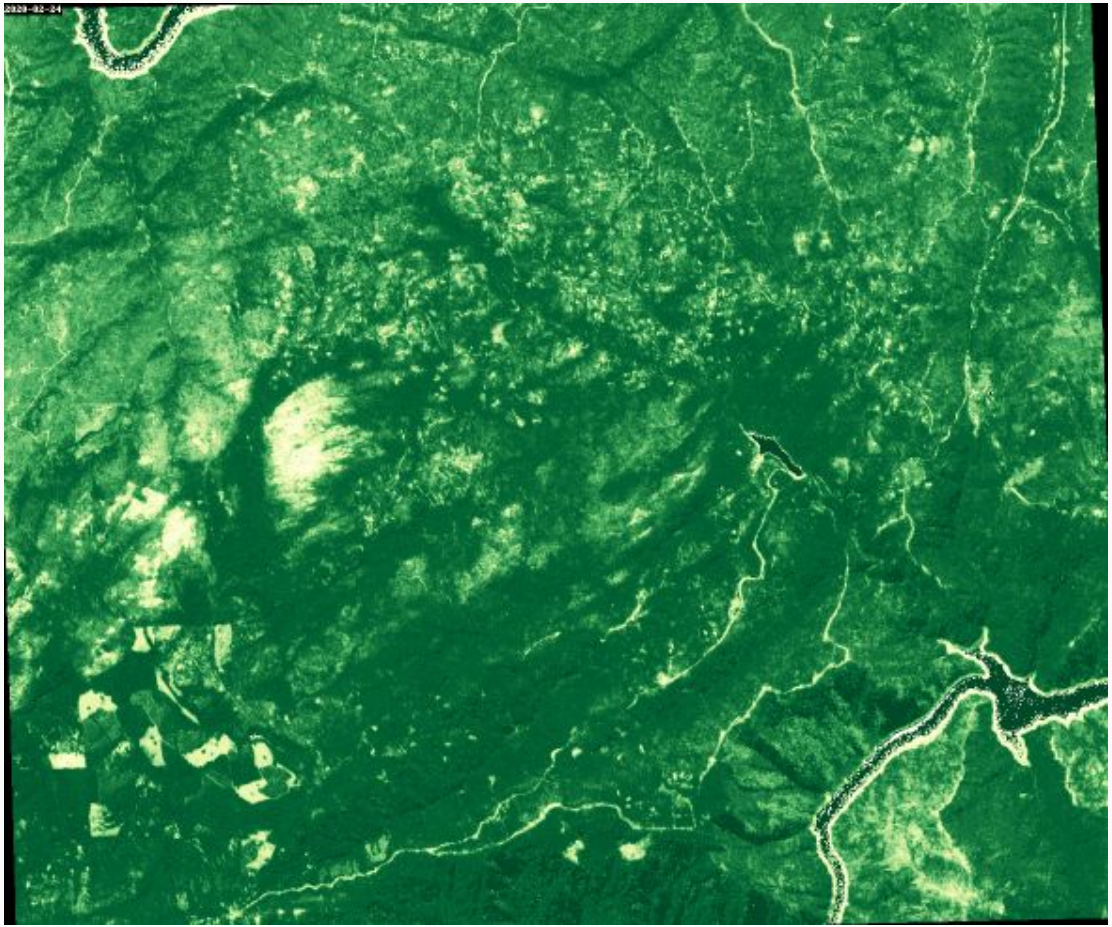



```
$ time sat-fetch load sentinel-scenes.geojson \
    --download red green blue
```

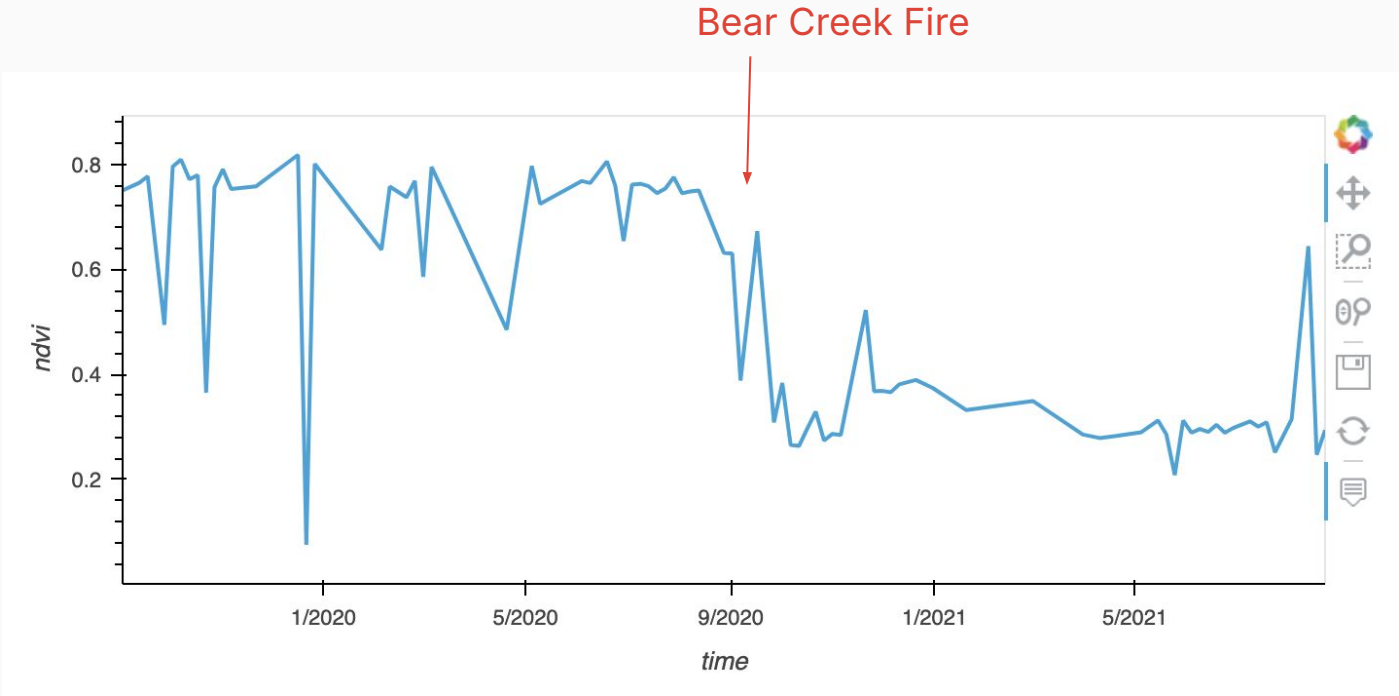
Planetary Computer - Bear Creek Fire 2019

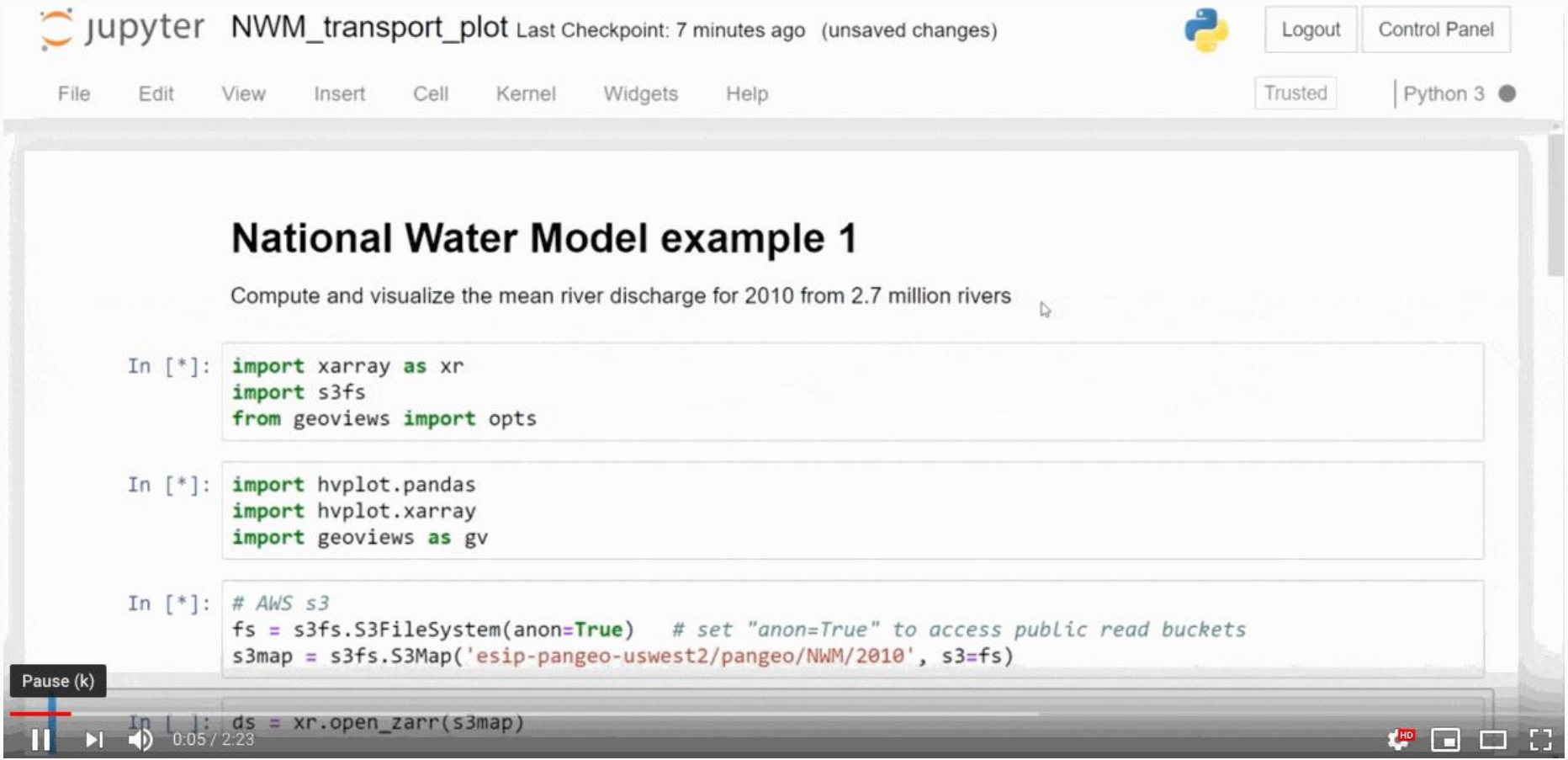



Status
Workers
Tasks
System
Profile
Graph
Groups
Info
More...
Bytes stored: 3.16 GiB
0.0
32.0 GiB
64.0 GiB
96.0 GiB
128.0 GiB
160.0 GiB
192.0 GiB
Bytes stored per worker
0.0
1.0 GiB
2.0 GiB
3.0 GiB
4.0 GiB
5.0 GiB
6.0 GiB
Processing
CPU
Occupancy
Tasks Processing
0
0.2
0.4
0.6
0.8
1
Task Stream
479000us
479020us
479040us
479060us
479080us
Progress



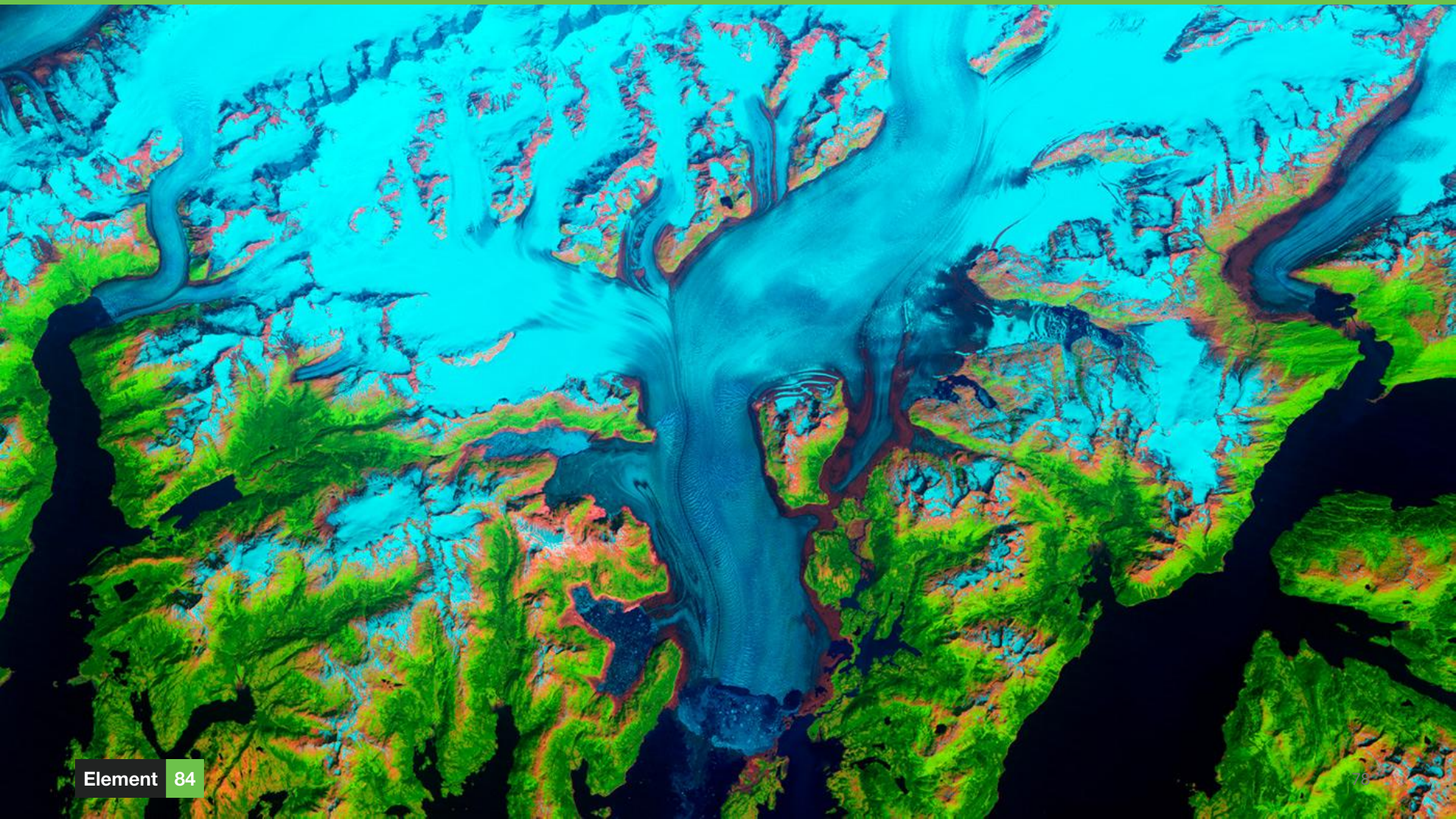
Two years of NDVI




jupyter NWM_transport_plot Last Checkpoint: 7 minutes ago (unsaved changes)
Logout
Control Panel
File Edit View Insert Cell Kernel Widgets Help
Trusted
Python 3
National Water Model example 1
Compute and visualize the mean river discharge for 2010 from 2.7 million rivers
In [*]:
import xarray as xr
import s3fs
from geoviews import opts
In [*]:
import hvplot.pandas
import hvplot.xarray
import geoviews as gv
In [*]:
AWS s3
fs = s3fs.S3FileSystem(anon=True) # set "anon=True" to access public read buckets
s3map = s3fs.S3Map('esip-pangeo-uswest2/pangeo/NWM/2010', s3=fs)
Pause (k)
In [ ]:
ds = xr.open_zarr(s3map)
0:05 / 2:23

Thank you!

@GeoSkeptic


Element 84